

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.Doi Number

Big Data and AI Algorithms for Sustainable Development Goals: A Topic Modeling Analysis

Prema Nedungadi^{1,2}, Simi Surendran¹, Kai-Yu Tang³, Raghu Raman⁴

¹School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri, Kerala, India

²AmritaCREATE, Amrita Vishwa Vidyapeetham, Amritapuri, Kerala, India

³Graduate Institute of Library & Information Science, National Chung Hsing University, Taoyuan City, Taiwan

⁴Amrita School of Business, Amrita Vishwa Vidyapeetham, Amritapuri, Kerala, India

Corresponding author: Author (e-mail: raghu@amrita.edu).

ABSTRACT This study makes significant contributions to the field by examining the transformative role of big data and artificial intelligence (AI) in advancing Sustainable Development Goals (SDGs), particularly healthcare (SDG3), sustainable energy (SDG7), and industry and infrastructure (SDG9). Using BERTopic modeling, a machine learning technique, this research systematically analyzes literature from 2013 to 2024, providing an overview of AI and big data applications mapped to SDGs which is a first. This structured approach identifies key SDGs impacted by these technologies and highlights interdisciplinary methods that further enhance SDG outcomes. AI applications notably improve healthcare by advancing disease tracking, tailored treatments, and precision medicine, fostering universal healthcare and reducing noncommunicable disease mortality. In energy, AI-driven solutions optimize forecasting, grid management, and renewable integration, while in industry, they bolster infrastructure resilience through innovations like predictive maintenance and automated quality control within Industry 4.0 frameworks. The integration of automated text analysis and semantic context captures broad trends, contributing both methodologically and substantively at the intersection of AI and sustainability. Despite these advancements, the study underscores ethical concerns, including data privacy, security, and algorithmic biases. Interdisciplinary collaboration among healthcare professionals, engineers, environmental scientists, and AI experts is crucial to developing ethical, scalable AI solutions. The study suggests future research focus on AI transparency, scaling across diverse sectors, and integrating advanced techniques such as neurosymbolic AI and quantum neural networks to enhance system reliability. These insights offer practical implications, reinforcing the potential of AI and big data to address global challenges sustainably while calling for balanced attention to ethical and regulatory dimensions.

INDEX TERMS Sustainable development goal, Big data, Artificial intelligence, Healthcare, Resilient energy, Resilient infrastructure, Industrial innovation, Generative AI

I. INTRODUCTION

The rapid advancement of artificial intelligence (AI) technologies has transformed our ability to address the global challenges of climate change, healthcare, and industrial inefficiencies. Sophisticated tools for analyzing vast datasets enable more efficient decision-making, driving innovation in sectors critical to achieving the United Nations Sustainable Development Goals (SDGs). The contemporary landscape of global scientific challenges, including climate change and public health crises, benefits significantly from the integration of big data and AI. These complex challenges require advanced

analytical tools for managing massive datasets and deriving actionable insights [1]. Big data and AI are crucial for progress toward the SDGs set by the United Nations, which address key issues such as inequality, environmental sustainability, and peace. They provide a robust capacity for managing vast amounts of information, including data-driven decisions, in tackling these multifaceted challenges [2][3][4][5].

This study aims to examine how big AI-driven data solutions have addressed scientific challenges related to the SDGs. The growth in global data heightens the complexity and scale of these challenges, necessitating scalable, effective solutions [6][7]. The relevance of big data and AI

extends across natural and life sciences, including biotechnology [8], education [9], geophysics [10,18], genomics [11,13], heritage conservation [12], and medicine, as well as engineering fields such as computer science, space technology [14], chemical engineering [15] and civil engineering [16,17], highlighting their versatility and critical role in diverse scientific inquiries.

AI and big data have driven advancements across numerous fields. Initially centered around data-intensive areas such as computer engineering and information systems, these technologies have enabled foundational methodologies for handling complex datasets [19][20]. As they have evolved, their applications have broadened, impacting diverse fields, from physics and urbanism to bioinformatics advancements [21][22] and sustainable urban planning [23][24][25]. Their growing applications have intensified their impact within fields and encouraged interdisciplinary approaches essential for complex problems, highlighting their dynamic capabilities [26][27].

AI and big data inherently promote an interdisciplinary approach, merging insights from various fields to address complex challenges. The future of research on these technologies is geared toward enhancing their ethical application, ensuring data security, and developing accessible solutions for disadvantaged regions [28][29][30]. The ongoing advancement of AI and big data is expected to address previously unsolvable problems in emerging fields such as space and mechanical engineering, with researchers continually exploring ways to improve the performance/complexity ratio of AI systems, aiming for sophisticated yet resource-efficient solutions [31][32]. This direction is vital for aligning innovations with sustainable development needs, effectively contributing to global challenges, and paving the way for a more resilient and equitable future.

The literature includes several studies exploring the intersection between AI and the SDGs. [33] analyzed this correlation, identified future research directions, and highlighted the roles of green AI and eco-friendly Internet of Things (IoT) applications in governance and data management for SDG achievement. AI's sustainability challenges from environmental and social perspectives can address issues such as carbon emissions and fairness [34]. [35] emphasized establishing responsible AI practices and addressing sustainability challenges, whereas [36] examined how AI and big data can engineer solutions for the SDGs. [37] discussed the potential of AI to reduce greenhouse gas emissions across various sectors, and [38] analyzed the impact of AI on industry, education, and sustainability efforts. [39] used predictive modeling to address environmental health and ecosystem issues.

The integration of big data and AI is crucial for addressing global challenges aligned with the SDGs. Despite their potential, significant gaps persist in

understanding their specific impacts across different SDGs and the effectiveness of interdisciplinary approaches.

To analyze the literature systematically, this study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol, ensuring a rigorous approach to data collection and analysis. Additionally, BERTopic modeling is used for identifying key themes within AI and big data research, providing deeper insights into their specific contributions to SDG-related fields. This study aims to address gaps in our understanding of the contributions of AI and big data to sustainable development by answering the following research questions:

RQ1: Which SDGs are most impacted by AI and big data technologies?

RQ2: What specific contributions have big data and AI made toward advancing these SDGs?

RQ3: How do interdisciplinary approaches enhance the achievement of these SDGs?

RQ4: What are the emerging trends and future research directions for big data and AI addressing these SDGs?

RQ5: What ethical, data security, and accessibility challenges arise with the deployment of big data and AI in addressing these SDGs, and what policies are recommended to address these issues?

This paper makes several significant contributions to the field. First, it applies BERTopic modeling, a machine learning technique, to systematically analyze literature published from 2013 to 2024. This approach provides a comprehensive overview of the current research landscape regarding the application of big data techniques and AI in emerging scientific fields mapped directly to SDGs, which is a first. Using this structured understanding of the research landscape, the study highlights which SDGs are most impacted by AI and big data, exploring how interdisciplinary approaches can enhance their achievement. Second, the study effectively integrates automated text analysis with semantic context to capture broad trends through topic modeling, thereby making methodological and substantive contributions at the intersection of AI and sustainability. Finally, the paper identifies emerging trends and offers practical implications for future interdisciplinary research directions concerning big data and AI applications in healthcare, energy management, and industrial innovations.

The structure of this paper is organized as follows: Section II reviews the literature and identifies the gaps. In Section III, we describe the systematic approach employed for data collection and analysis. Section IV presents the key findings from BERTopic modeling. In Section V, we present various applications and emerging trends of AI concerning specific SDGs, interpreting our findings in depth. Finally, the conclusion summarizes the main takeaways and explores their implications for stakeholders.

II. RELATED WORK

The convergence of AI and big data analytics has emerged as a promising paradigm for providing opportunities and challenges across multiple sectors, spanning healthcare, education, environmental conservation, sustainable energy, and industrial innovation. In recent years, there has been considerable scholarly interest in investigating the impact of AI on the United Nations SDGs. Many review papers have emerged that aim to consolidate the literature in this domain to provide insights into the utilization of AI for advancing the objectives of sustainable development.

A prevailing thematic focus observed is the role of AI in improving environmental sustainability, waste management, water management, CO₂ emission reduction, carbon foot printing, and climate change mitigation are key areas where AI interventions are extensively examined. One study examined machine learning approaches that can address the sustainability challenges associated with AI from two key perspectives: environmental and social sustainability [34]. The authors addressed various issues, including carbon emissions from AI model training and inference, the substantial resources required for data collection and annotation, concerns regarding fairness and privacy in AI models, and ensuring the safety of AI against adversarial attacks and data poisoning. They categorize AI technologies into four main groups: computation-efficient AI, data-efficient AI, responsible AI, and resilient AI. Computation-efficient AI focuses on reducing the environmental impact of AI by developing compressed AI models and applying techniques such as pruning. Data-efficient AI, including transfer learning and active learning algorithms, promotes sustainable development by reducing manual efforts and resources. Responsible AI emphasizes considering AI ethics in system design, particularly fairness and user data privacy. Resilient AI highlights the importance of rationalizing AI models to enhance social sustainability and acceptance of modern AI systems.

[35] stated that AI systems have been developed to mitigate societal, ecological, and economic issues. They examined various aspects, including establishing rules for responsible AI and emphasizing transparent and reasonable responsibilities for developers, operators, and decision-makers. They formulated sustainability challenges for AI, questioning the suitability of commonly used datasets for transforming existing productions and exploring the democratic legitimization of AI-supported sustainability efforts. It also scrutinizes the resource and energy intensities of AI, evaluating whether the benefits of AI justify the energy and emissions generated during its lifecycle. Moreover, the review targeted the regulation of market power and monopolies in the AI sector, addressing revenue generation and data concentration, and proposed regulatory frameworks to safeguard consumer interests and ensure liability accountability.

The role of AI in reducing greenhouse gas emissions and adapting to environmental challenges was discussed in [37]. The authors examined studies that employed AI in various sectors, such as electricity, forest fire management, transportation, and carbon footprint tracking. They highlighted the potential of machine learning algorithms, including linear regression and support vector machines, as well as deep learning approaches, like convolutional neural networks, in tackling climate change challenges. Additionally, they deliberated on the advantages, pitfalls, and prospects of integrating AI with environmental challenges, emphasizing the need for technological solutions to address these issues. In addition, a review by [38] examined the influence of AI on the SDGs, investigating its applications in education, healthcare, agriculture, and climate change mitigation. The potential of AI to revolutionize industry, education, and sustainability efforts by enabling smarter interventions, reducing waste, creating new applications, and improving accessibility was also discussed. Moreover, they introduced a sustainable AI framework aligned with the SDGs to enhance the outcomes in these areas.

On the other hand, [39] surveyed the practicality of human reasoning approaches accessible to beginners, who may lack expertise in computer programming or advanced mathematics, for data analysis and result interpretation. Additionally, the authors investigated the availability of user-friendly AI-based software for beginners, facilitating experimentation. They utilized the 2018 Environmental Performance Index (EPI) data from 180 countries, encompassing indicators of environmental health and ecosystem vitality. The system employs an AI-based Bayesian network approach for predictive modeling to uncover underlying tensions between environmental health, which typically improves with economic growth and rising affluence, and ecosystem vitality, which often deteriorates because of industrialization and urbanization.

[40] explored recent advancements in AI and deep learning (DL) for achieving the SDGs, focusing on renewable energy, environmental health, and smart building energy management. In the renewable energy sector, AI and DL techniques have been shown to optimize energy management, fault detection, and power grid stability. The study presented various approaches, including deep reinforcement learning for optimizing AC power flow, clustering techniques for solar power generation, and predictive models such as SVR, AdaBoost, and deep neural networks for renewable energy systems. Additionally, this paper examines the utilization of GeoAI for the efficient processing of spatial data and the potential of deep reinforcement learning in smart building energy management. The authors also discussed digital twin-based methods proposed for intelligent optimization and automation systems in residential energy management. While existing research review papers have provided

insights into the broad applications of AI and big data in various sectors, this review distinguishes itself by focusing specifically on the techniques and algorithms used in application areas of the top three researched SDGs. Focusing on the top three SDGs, this paper offers a targeted examination of how AI and big data can contribute to achieving specific sustainable development objectives. Additionally, this study suggests future research directions in these fields and recommends AI algorithms to address challenges associated with various SDG-related use cases.

III. METHODOLOGY

In alignment with the PRISMA guidelines, this research followed a structured protocol for the identification, screening, eligibility assessment, and inclusion of publications. The robustness and reliability of the PRISMA protocol are well established, and it has been utilized in systematic literature reviews across various domains [41][42]. Figure 1 illustrates the research design.

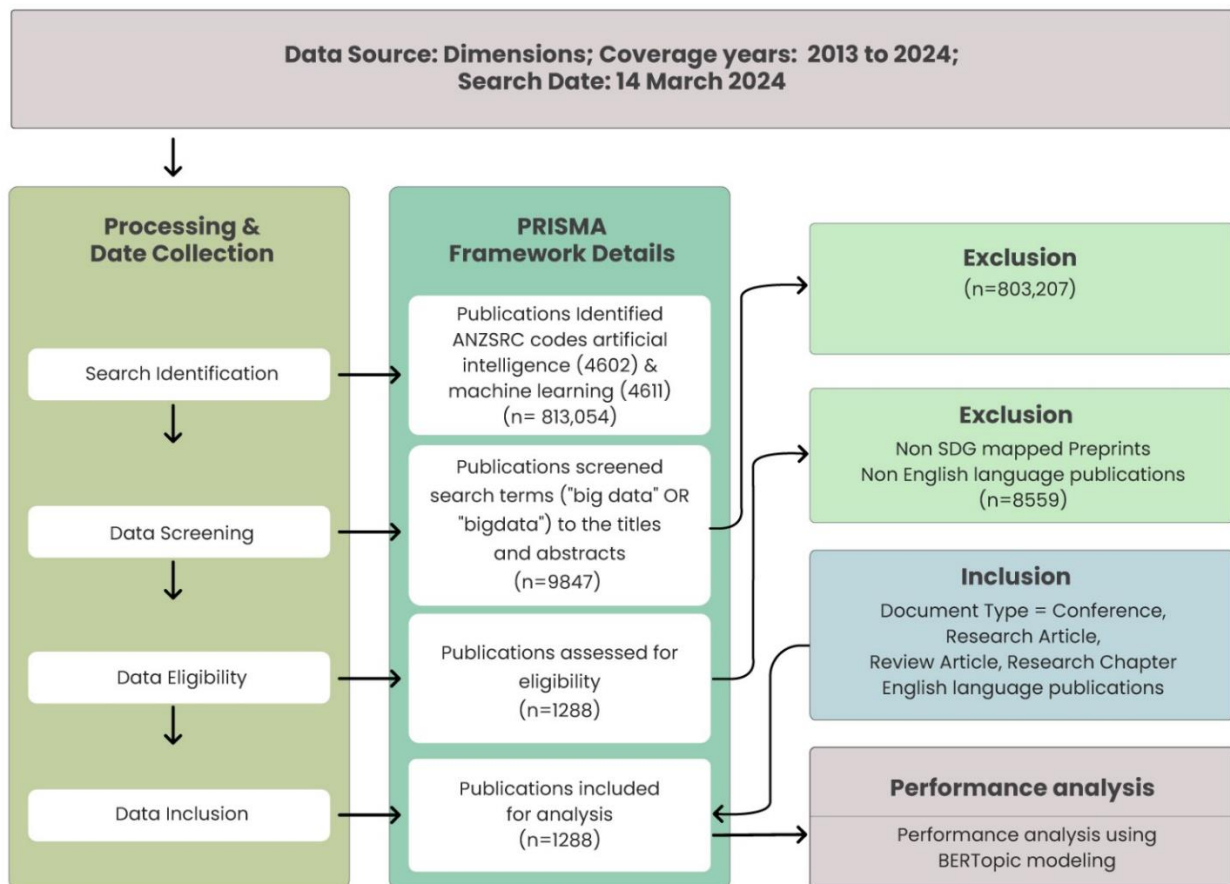


FIGURE 1. Research design

A. IDENTIFICATION PHASE

Employing the Dimensions database, which has 82.2% journal coverage beyond the Web of Science database and 48.1% journal coverage beyond the Scopus database, this study used an exhaustive search of artificial intelligence and machine learning research, utilizing ANZSRC codes 4602 and 4611, respectively [43]. The Dimensions database's application of the ANZSCO framework for categorizing research fields facilitated the identification of 813,054 publications from 2013-2023 [44].

B. SCREENING PHASE

During the screening, we applied search terms ("big data" OR "bigdata") to the titles and abstracts, which resulted in 9847 publications.

C. ELIGIBILITY PHASE

A focused selection of publications mapped directly to the 17 SDGs was conducted via proprietary algorithms within the Dimensions database, influenced by various SDG mapping initiatives, including the Aurora Network-Global's SDG-Queries, the University of Auckland's methodology (Auckland SDG mapping, n.d.), and Elsevier's SDG Mapping Initiatives (Elsevier's, n.d.). The Digital Science SDG Mapping Initiative was selected for seamless integration with Dimensions, providing preset search queries for each SDG, supported by a machine-learning model refined through expert review. Various studies have utilized detailed SDG mapping

methods to examine the trajectory of research themes. For example, the area of green hydrogen was scrutinized by [44], while the subjects of Fake News and the Dark Web were investigated by [41] and [42], respectively. These findings illustrate the use of SDG mapping as a tool to gauge how academic endeavors align with and influence sustainable development goals.

Our approach acknowledges the interconnected nature of the SDGs, a complexity that network analysis literature, such as [44] and [45], has previously characterized. The use of co-occurrence maps illustrates the semantic proximity of SDGs, revealing conceptual relatedness as indicated by shared citations in the literature [46]. The produced map visualizes each SDG as a node, with the node's size reflecting the extent of its presence in the research literature. The thickness of the lines connecting these nodes signifies the frequency of co-occurrence among the goals, thus illustrating the interconnectedness of the SDGs. This depiction reveals the SDGs as an intricate but unified framework that underpins efforts toward global sustainability [47]. To ensure the quality and reliability of the data, English-language research articles, review articles, conferences, and research chapters were included, and preprints were excluded. This phase resulted in 1288 eligible publications. The VOSviewer application maps and interprets the research landscape in the context of SDGs [48].

D. INCLUSION PHASE

All 1288 eligible publications were included in the final dataset for detailed analysis.

E. BERTopic MODELING

Topic modeling techniques, such as nonnegative matrix factorization (NMF), latent Dirichlet allocation (LDA), probabilistic latent semantic analysis (PLSA), and To2Vec, often struggle to capture semantic word relationships and are limited in handling short texts [49]. Unlike traditional bag-of-words methods that prioritize term frequency, BERTopic employs embeddings [50] to represent documents in a lower-dimensional space, thereby preserving semantic information and providing a more contextual understanding [51] [45]. BERT, a deep learning language model developed by Google [52], leverages bidirectional encoder representations from transformers to enhance semantic understanding.

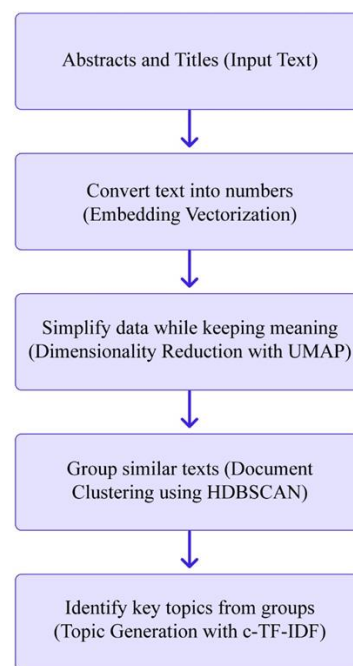


FIGURE 2. BERTopic modeling workflow

As shown in Figure 2, the initial step in our modeling process involves embedding vectorization, which transforms the input text into numerical representations known as embeddings. Subsequently, dimensionality reduction is achieved via Unified Manifold Approximation and Projection (UMAP), which effectively groups similar data points, leading to more distinct topic clusters [53]. After simplifying the data, hierarchical density-based spatial clustering of applications with noise (HDBSCAN) is employed to identify clusters of closely packed data points, disregarding scattered or irrelevant information. To interpret the meaning of each topic cluster, the class-based term frequency-inverse document frequency (c-TF-IDF) is utilized to extract the most salient words or phrases, facilitating the identification and ranking of key topics within the documents [54]. For our analysis, we adopted the "all-MiniLM-L6-v2" text representation model, which is well suited for clustering and semantic search tasks in English [55]. This modified TF-IDF approach identifies representative terms within each topic by considering both term frequency and inverse document frequency. By focusing on terms that appear frequently within a document but less frequently in others, the model can effectively extract key concepts. These terms are then used to assign topics to documents, with associated probabilities indicating the likelihood of document membership within each topic [56].

To enhance topic model performance, we carefully tuned three key hyperparameters: the n-gram range, the number of topics, the minimum topic size, and the number of keywords per topic. By setting the n-gram range to (1, 2), we consider

both single words and two-word phrases, enabling a more comprehensive understanding of contextual relationships without excessive complexity. The number of topics was iteratively adjusted between 5 and 20, and the intertopic distance and coherence scores were evaluated to achieve a balance of specificity and relevance. The minimum topic size was set between 20 and the number of keywords per topic, ensuring the exclusion of overly granular or irrelevant topics by requiring a sufficient number of documents for each topic. Simultaneously, the number of keywords per topic was carefully selected to capture the most relevant terms, avoiding excessive detail. These settings, as emphasized by [93], effectively balance specificity and usability, resulting in meaningful and practical topics for analysis. Following stopword removal, common words such as 'use', 'used', 'use', 'add', 'added', 'adding', 'useful', 'based', 'related', and others were eliminated.

The UMAP algorithm parameters were configured to their default settings, with "calculate probabilities" enabled to assess the likelihood of document–topic associations and the language specified in English. The number of topics was determined through a combination of quantitative and qualitative evaluations, emphasizing intertopic distance and coherence scores. We experiment with topic numbers ranging from 4-20, ensuring a minimum distance of 0.05 to maintain sufficient separation between points in the reduced-dimensional space. The "cosine" metric was used to calculate the angular similarity between vectors, and a random state of 100 was employed for reproducibility. The nearest neighbor ($n_neighbors$) parameter was set to 15, with a focus on the local neighborhood of the data points to preserve the detailed structure while capturing broader patterns [57].

Although machine learning techniques excel at clustering data, the risk of misclassification persists [58]. To validate and interpret our results, we conducted a comprehensive manual review of both the identified topics and their representative publications. A team of three experts meticulously evaluated the topics qualitatively, ensuring their coherence and meaningfulness, aligning with recent BERTopic studies [59][60]. The experts thoroughly

examined topic keywords and representative publications and analyzed abstracts, titles, and, when necessary, full papers. Additionally, probability values and citation counts were used to select the top three representative articles for each topic. This rigorous review process guarantees the accuracy and utility of our unsupervised topic modeling results, facilitating the identification of key themes within the data.

Compared with the use of AI and big data analysis, the features of BERTopic modeling used in this paper are summarized. First, we use the pretrained BERT language model as the backbone to understanding the semantic context instead of random word combination results from the AI tool. Second, the method combines advanced machine learning techniques such as UMAP, HDBSCAN, and topic models to optimize topic consistency and interpretability. Third, we use the enhanced c-TF-IDF for topic keyword extraction, which considers both word frequency and word uniqueness. In addition, we employ a rigorous manual review of topics and representative literature to ensure the reliability of the results. This positions the research to make meaningful methodological and substantive contributions at the intersection of data science and sustainability.

IV. RESULTS

A. EVOLUTION OF BIG DATA AND ARTIFICIAL INTELLIGENCE ACROSS FIELDS OF RESEARCH

Figure 3 shows the influence of big data and AI across various scientific fields over the years. Grounded in communications engineering, mathematical sciences, computer vision, and information systems, these domains set the foundational data and computational techniques crucial for developing big data and AI. As the timeline advances, there is an evident integration of big data and AI in fields such as software engineering, highlighting the need for software capable of managing and interpreting vast datasets. The emergence of linguistics and data management indicates an increased focus on processing natural language and structuring data core elements in AI developments.

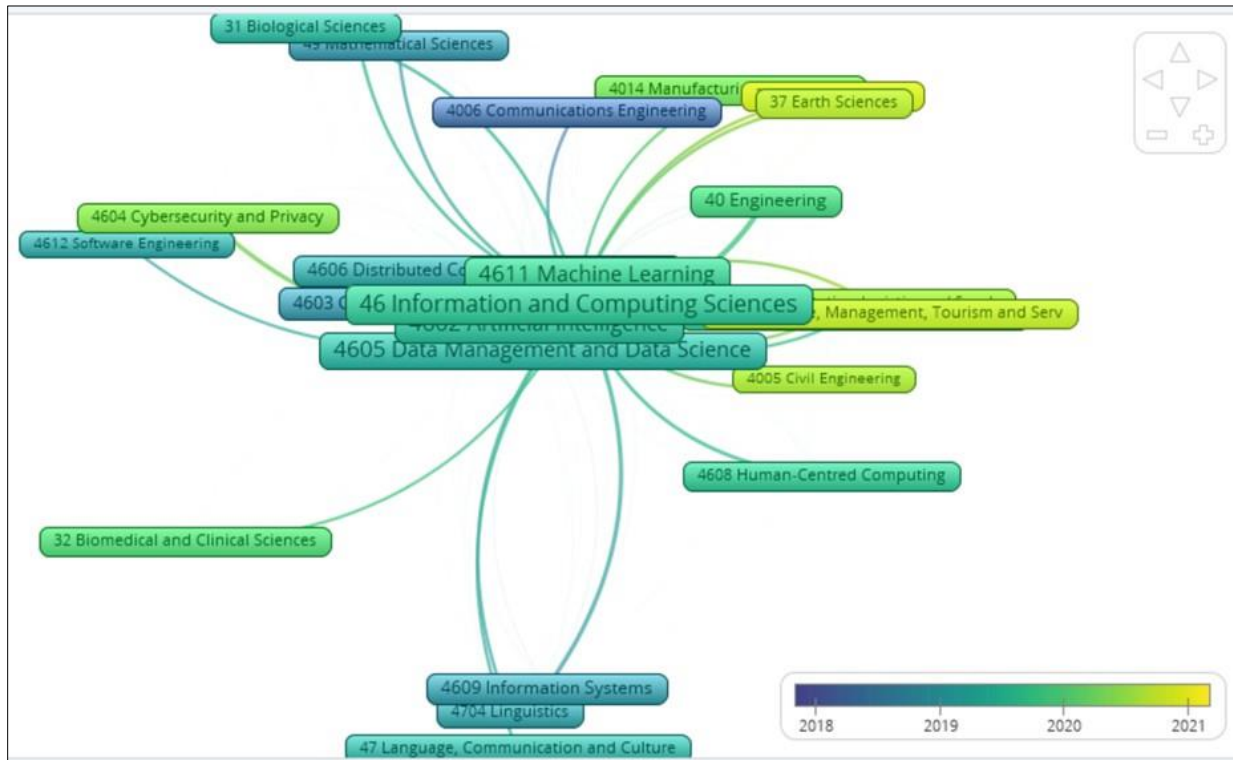


FIGURE 3. Evolution of big data and AI across diverse research disciplines.

The growing prominence of biomedical and clinical sciences signals the intersection of big data with life sciences, where AI algorithms are increasingly employed to analyze complex biological information. Machine learning is essential in predictive analytics and data pattern recognition, which are fundamental to big data applications. In the latter part of the timeline, the rise of cybersecurity and privacy research points to the importance of AI in protecting data integrity and addressing privacy concerns within massive datasets. Civil engineering and earth science also reflect the use of AI and big data in modeling and solving environmental challenges, optimizing resource use, and improving urban planning. In commerce, management, tourism, and services, the implications of big data and AI influence customer behavior analysis, market trend prediction, and service personalization. Finally, geoinformatics uses AI to interpret geographic data, providing insights into spatial patterns and relationships crucial for decision-making in various sectors.

B. MOST-IMPACTED SDGs THROUGH BIG DATA AND AI INNOVATIONS (RQ1)

Next, we analyze how big data and AI have influenced global scientific challenges through the lens of interdisciplinary SDGs. Starting with "Life on Land", the emphasis indicates a growing concern for biodiversity and ecosystem services, which big data and AI address through enhanced monitoring and analysis of environmental data

[61][62][63]. "Sustainable Cities and Communities" follows, reflecting a focus on urban sustainability where AI-driven data analytics help optimize resource use and urban planning [64][65][24]. By 2020, "climate action" and "affordable and clean energy" leveraged AI for climate modeling, renewable energy optimization, and sustainable energy solutions [66]. The progression toward "reduced inequalities" suggests the application of these technologies in creating equitable solutions, such as through predictive analytics, to serve underserved communities better.

"Quality education" includes the adoption of AI in personalized learning and educational data mining [67][68]. The sequence continues with "Industry, Innovation, and Infrastructure," indicating an interplay where AI and big data are critical in driving innovation, improving infrastructure, and promoting industrial sustainability [69]. By 2020, the focus shifted to "good health and well-being," highlighting AI's role in healthcare through data-driven diagnostics, personalized treatment, and managing health crises [70]. In 2021, the attention given to "Zero Hunger" reflected the use of AI in agricultural technologies to enhance food security, whereas "Responsible Consumption and Production" highlighted the culmination of AI and big data in creating sustainable consumption patterns and improving production efficiency [71].

Figure 4 shows how big data and AI research are instrumental in tackling complex global challenges and driving progress toward a sustainable future. The figure

indicates that among the SDGs, “good health and well-being” has likely been the most impacted by AI and big data techniques in recent years. Innovations in diagnostics, personalized medicine, and patient care have led to improved health outcomes. Additionally, AI-driven tools and telemedicine have expanded access to healthcare services, particularly in remote areas, making a significant difference in developing countries. Following SDG3, SDG9 is the second most affected by AI. It enhances innovation in industries by optimizing processes, improving efficiency, and driving new product development. AI also supports smart infrastructure through predictive maintenance and advanced manufacturing techniques, promoting sustainable industrial growth. SDG7 is the next group in which AI contributes to optimizing energy consumption, enhancing grid management, and advancing renewable energy technologies. Although AI has a role in the

advancement of SDGs 7 and 9, its direct impact on health makes SDG3 the most significant.

C. BERTopic THEMES FROM SDG3, SDG7 AND SDG9

Next, we explored major themes from the top three SDGs, SDG3, SDG7, and SDG9, via BERTopic modeling (Figure 5). The three topics in SDG3 are AI-enhanced diagnosis, deep learning with health data, and AI for disease prediction. SDG7 has four topics: optimizing energy systems using AI, AI-driven models for energy forecasting, sustainable energy efficiency, and optimization based on AI algorithms. The four topics in SDG9 are leveraging AI and big data to transform industrial infrastructure, AI-driven sustainable industrial innovation, intelligent manufacturing for Industry 4.0, and predictive maintenance.



FIGURE 4. Progression of big data and AI research toward various sustainable development goals.

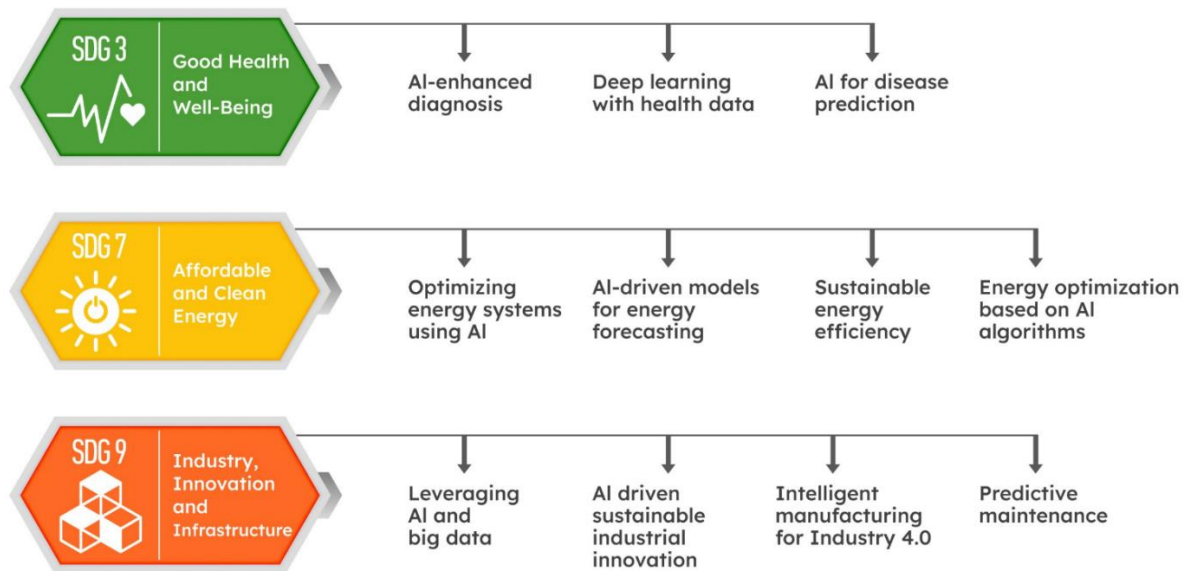


FIGURE 5. Evolution of big data and AI across diverse research disciplines

D. SDG3: IMPACT ON GLOBAL HEALTH (RQ2)

1) AI-ENHANCED DIAGNOSIS

To achieve SDG3, many AI and big data tools are being used to improve diagnosis. By analyzing large volumes of data, such as health records, medical images, and genetic information, AI systems can find patterns. This analysis allows for quick diagnosis and decision-making, which can lead to better outcomes for patients. [72] integrated Apache Spark with machine learning to enhance the early diagnosis of chronic kidney disease, leveraging feature selection methods and multiple classifiers to achieve high accuracy. Such research supports preemptive health interventions, aligning with the SDG3.4 target to reduce premature mortality from non-communicable diseases. The study by [73] leverages big data from electronic medical records to enhance medical diagnostics through deep Q-learning coupled with gorilla troop optimization to select features for classifying breast cancer cases. This integration aligns with SDG3.8 by promoting health and well-being through advanced diagnostic tools and improving patient care through technological empowerment. [74] explored the application of AI in mental health care, proposing techniques such as AI-driven virtual assistants and chatbots to offer evidence-based recommendations for personalized treatments. They suggested machine learning tools to analyze electronic medical records, illustrating AI's ability to increase diagnostic accuracy and treatment efficacy in orthopedics, thereby contributing to the broader goals of SDG3.8 by improving health outcomes and medical procedures.

2) DEEP LEARNING WITH HEALTH DATA

Deep learning models can automatically learn and extract features to analyze medical data such as X-ray, MRI,

and CT data to detect abnormalities with high accuracy. Because these models can manage large volumes of data, they can easily scale to accommodate today's increasing data availability. [75] reviewed how healthcare services are improved by the use of robust processors, machine learning, and deep learning algorithms facilitated by big data and artificial intelligence. This review exemplifies the integration of sophisticated AI techniques to manage healthcare data, directly supporting SDG3.d by improving global health security and disease surveillance systems. [76] introduced a novel approach using deep similarity learning to increase the predictive accuracy of electronic health records. Their method, which employs CNNs within a Siamese-based framework, addresses data sparsity and heterogeneity in medical big data. The technique exemplifies the potential of AI in personalized patient care, thereby supporting SDG3.4's aim of ensuring healthy lives through innovative and effective health solutions. [77] utilized AI to revolutionize dermatological diagnostics. By employing pretrained CNN models through transfer learning, they achieved high accuracy in skin disease classification. By harnessing electronic health records, patient monitoring devices, and smartphones, they advocated the use of big data as a service framework to identify skin condition patterns to facilitate targeted preventive measures. This approach enhances diagnostic efficiency and reduces the likelihood of human error, supporting SDG3.8 by improving the quality of health services and ensuring access to essential healthcare technologies.

3) AI FOR DISEASE PREDICTION

AI-assisted disease forecasting enables healthcare practitioners to make timely interventions, develop

personalized treatment plans, and improve patient health outcomes. [78] focused on the use of artificial intelligence and big data technologies in traditional Chinese medicine orthopedics to optimize rehabilitation information and demonstrated that such systems can be faster and more convenient. This technological advancement is set against the backdrop of creating a "health cloud platform" to integrate comprehensive community health services. Machine learning algorithms such as support vector machines are utilized within the orthopedic rehabilitation information data mining system to categorize different types of diseases. This approach directly supports SDG3.8, which is about achieving universal health coverage, including access to quality essential healthcare services. [79] used the Cleveland dataset to predict heart disease at an early stage via the Jellyfish optimization algorithm for feature extraction. The resulting optimal feature set was then employed in machine learning algorithms, including artificial neural networks (ANNs), decision trees, support vector machines (SVMs), and AdaBoost, for predicting heart abnormality treatment planning, directly supporting the SDG3.8 goal of promoting well-being through better disease management. A study by [80] introduced an automated system that mimics an initial doctor's diagnosis via machine learning to analyze symptoms and recommend precautions, achieving nearly perfect accuracy. This development helps extend healthcare resources and enhances patient education on disease management, contributing to the aim of SDG3.4 for universal health coverage.

E. SDG7: IMPACT ON AFFORDABLE AND CLEAN ENERGY (RQ2)

1) OPTIMIZING ENERGY SYSTEMS VIA AI AND BIG DATA TECHNIQUES

In the research [81], artificial intelligence (AI) and big data techniques were applied to optimize the extraction of oil and gas, a key component of the energy sector. The utilization of neural networks and evolutionary algorithms enhances the extraction process, ensuring more efficient energy production. This aligns closely with SDG7's aim for enhanced energy efficiency (SDG7.3) and renewable energy sources, as better extraction techniques can reduce the environmental impact of fossil fuel extraction and provide more reliable energy sources. [82] focused on improving the reliability and efficiency of smart grids via AI-driven analytics to correct data outliers. This is pivotal for SDG7, as enhancing grid efficiency directly contributes to the availability of affordable and clean energy (SDG7.1). The distributed algorithm proposed in this research uses the alternating direction method of multipliers for imbalanced data classification, which saves training time and improves scalability. By ensuring that energy distribution is managed more efficiently through sophisticated AI algorithms, this research helps reduce energy waste and optimize

distribution systems. The exploration of how big data and AI can significantly reduce energy consumption within smart buildings was highlighted in a study by [83]. The integration of machine learning techniques for energy management aligns with SDG7 by promoting energy efficiency in urban environments (SDG7.3). The authors developed a data fusion framework to process power grid big data generated from heterogeneous sources via imprecise reasoning and clustering algorithms. Reducing energy consumption conserves energy and sets a framework for sustainable urban development, which is crucial as urban areas continue to grow and consume more resources.

2) AI-DRIVEN MODELS FOR ENERGY FORECASTING

AI-based energy forecasting systems can analyze data on energy consumption and market trends to predict future energy demands and ensure a reliable supply for all consumers. The use of long short-term memory (LSTM) networks combined with human behavior pattern recognition by [40] significantly enhanced the prediction and management of home energy consumption. This study leveraged data from residential customers to forecast loads, incorporating LSTM with human behavior pattern recognition and multilayer neural networks. By improving load forecasting methods, this research supports the SDG7.3 subtarget to double the global rate of improvement in energy efficiency. These advancements facilitate more informed and efficient energy use, contributing to overall reductions in energy consumption. [84] leveraged deep learning to forecast wind power generation accurately, facilitating the integration of renewable energy sources into power grids aligned with SDG7.2, which discusses substantially increasing the share of renewable energy in the global energy mix. By enhancing the prediction accuracy through principal component analysis-based dimensionality reduction and support vector machines, this research contributes to improving the planning and utilization of wind energy and advancing sustainable energy solutions. By applying machine learning to predict energy consumption efficiently, [85] addressed the challenge of managing large datasets within smart grids. Leveraging H2O's distributed deep neural network implementation with support vector regression, the proposed approach not only helps reduce the computational load but also enhances the precision of energy consumption forecasts. This directly contributes to SDG7, a subtarget for enhancing international cooperation to facilitate access to clean energy research and technology, including renewable energy and energy efficiency.

3) SUSTAINABLE ENERGY EFFICIENCY THROUGH AI AND BIG DATA

[86] explored energy-efficient hardware designs for AI and machine learning, focusing on approximate computing to improve the power and energy efficiency of data processing systems. This approach directly supports SDG7's subtarget of enhancing global energy efficiency

(SDG7.3), as it reduces the energy demand of AI systems. By designing hardware that requires less energy, this paper contributes to sustainable technological advancements that can process large datasets without a proportional increase in energy consumption. [87] reviewed energy-efficient neural network implementations via GPUs and emerging memristor technologies. This work is crucial for SDG7, particularly for enhancing the efficiency of big data operations and reducing the energy footprint of processing large-scale neural networks (SDG7.3). This contributes to more sustainable AI applications by significantly lowering the power required for data analytics, which is essential for energy-intensive sectors. [88] reviewed accelerators for deep neural networks, addressing the challenges posed by big data and AI, such as speed, memory requirements, and energy efficiency. Optimizing the design of deep neural network accelerators via ASICs and FPGAs improved computational efficiency and reduced the latency and power consumption of AI systems. This research aids in achieving the SDG7 subtarget of increasing the share and efficiency of renewable energy by enabling faster and more efficient processing for AI-driven energy applications (SDG7.a).

4) ENERGY OPTIMIZATION ALGORITHMS IN BIG DATA APPLICATIONS

In the Internet of Everything era, managing exploding data volumes requires innovative solutions. [89] address the challenges posed by the Industrial Internet of Things, where devices often lack sufficient computing power and energy storage. By integrating mobile edge computing and deep reinforcement learning, their multiagent system optimally manages task offloading to conserve energy while reducing delay, directly contributing to SDG7.3, focusing on energy efficiency, and demonstrating the potent fusion of big data techniques and AI in industrial applications. [90] introduced a novel deep deterministic policy gradient-based algorithm tailored for 5G/6G communications to optimize the use of resources such as frequency and energy. This approach supports SDG7.3 by promoting energy efficiency and aligns with the overall theme of leveraging big data and AI to ensure sustainable energy management in next-generation communication systems. [91] explored the optimization of task migration in mobile edge computing environments to enhance the user experience while minimizing energy use and processing time. The use of the K-proximal policy optimization algorithm, an advanced form of deep reinforcement learning, exemplifies the application of AI to achieve greater energy efficiency, supporting the SDG7.3 subtarget. This research is pivotal in revealing how big data and AI can coalesce to optimize computational tasks in energy-constrained scenarios.

F. SDG9: IMPACT ON INDUSTRY, INNOVATION AND INFRASTRUCTURE (RQ2)

1) LEVERAGING AI AND BIG DATA TO TRANSFORM INDUSTRIAL INFRASTRUCTURE

A study by [92] reviewed how deep learning can monitor machine health by analyzing images and videos that capture machine health components. This vision-based predictive maintenance aids in the early detection and diagnosis of issues, ultimately minimizing downtime and enhancing overall machine efficiency. Their approach aligns with SDG9's focus on building resilient infrastructure (SDG9.1) and underscores the importance of innovation in achieving sustainable industrial processes (SDG9.5). [93] focused on optimizing network efficiency in industrial applications through adaptive clustering and machine learning, addressing SDG9 by improving the reliability and sustainability of industrial infrastructures (SDG9.4). Their methodology reflects an innovative approach to managing complex systems, ensuring operational efficiency and robustness, thereby fostering industry innovation and infrastructure (SDG9.5). In exploring next-generation wireless networks, [94] utilized AI and big data to enhance network capabilities, which are essential for modern industrial environments. This review examines the key factors influencing the adoption of big data analytics and the importance of advanced data analytics, ML, and AI in optimizing the operation, control, and efficiency of data-driven next-generation wireless networks.

2) AI-DRIVEN SUSTAINABLE INDUSTRIAL INNOVATION

By discussing the integration of big data and AI technologies, industrial cyber-physical systems can be enhanced through the use of machine learning and multiagent systems. This automation and efficiency are crucial for sustainable industrialization, aligning with the SDG9.5 target to foster innovation in resilient infrastructure. [95] proposed a traffic network flow prediction system that leverages a deep convolutional neural network alongside a parallel training algorithm for parameter learning. They utilized Spark cloud deployment for flexible scaling of computing resources and enhancing capabilities in processing large volumes of traffic big data. This discussion aligns with the SDG9.5 target to foster innovation in resilient infrastructures. [96] explored how the IoT and edge computing drove forward smart manufacturing systems. By integrating deep learning and collaborative robotics, their research promotes intelligent and sustainable industries, directly supporting SDG9.4 objectives to enhance sustainable industrialization and infrastructure. In the research highlighted by [97], a crucial infrastructure challenge concerning road damage detection was addressed by employing a deep learning approach based on YOLO. They collected data through a smartphone-based method and devised a lightweight, rapid solution for object detection tasks. The application of artificial intelligence within these frameworks contributes significantly to building robust infrastructures, aligning

with the robust infrastructure and innovation goals of SDG9.1 and SDG9.5.

3) INTELLIGENT MANUFACTURING IN INDUSTRY 4.0

The integration of AI with IoT and robotics provides machines with intelligence to optimize operations, manage failures, maintain quality control, and improve productivity. [98] delved into integrating deep learning in the context of road traffic to investigate traffic congestion patterns and explain nonrecurring congestion resulting from various events. They tested the performance of their model through three scenarios utilizing real-world data encompassing three event types: football games, hockey games, and traffic incidents. It emphasizes the role of AI in enhancing infrastructure, which directly supports SDG9.5 by fostering innovation in industrial solutions. [99] explored the role of computational intelligence in smart manufacturing, a core component of Industry 4.0. Their research, which uses the IoT and advanced AI algorithms such as deep learning and reinforcement learning, aligns with SDG9.4 by promoting sustainable industrialization. This approach improves manufacturing efficiency and resource management, directly contributing to smarter and more sustainable industrial practices. Focusing on the intelligent automation and operation management of large enterprise cloud data centers, [100] highlighted the use of big data and AI. This enhances the operational efficiency of data centers, which is fundamental for supporting sustainable industrial practices and infrastructure development. Such advancements contribute to the targets set by SDG9.1 and SDG9.4, promoting resilient and sustainable infrastructure.

4) PREDICTIVE MAINTENANCE

[101] developed a system that enhances defect detection in metal manufacturing through convolutional neural networks (CNNs) and extensive data mining. The authors proposed the use of machine learning and data fusion techniques to analyze manufacturing datasets to detect quality events and enhance process monitoring. They approached defect detection as a binary classification task and implemented ensemble learning on datasets derived from manufacturing processes. This technological integration improves product quality and optimizes resource use, supporting SDG9's focus on building resilient infrastructure and fostering innovation in industry processes (SDG9.4 and SDG9.5). [102] utilized deep convolutional neural networks to establish an adaptable robotic inspection station. This station is proficient in autonomously conducting quality control tasks within human-machine interface (HMI) consoles without human intervention. The tasks performed include operator recognition, classifying the type of HMI being inspected, and identifying errors in the display. [103] aimed to employ machine learning algorithms to uncover confusing patterns in IC testing data to increase quality and reduce the need

for manual labor. The application of predictive analytics exemplifies the transformative power of AI in Industry 4.0 to enable more efficient production lines and improve industrial sustainability, directly contributing to SDG9 by enhancing sustainable industrialization and infrastructure (SDG9.4).

V. DISCUSSION

This research examines the role of big data and AI in advancing the areas of healthcare, sustainable energy, industry, innovation, and infrastructure. AI techniques accelerate progress in SDG3 by enhancing services such as disease tracking, predicting disease progression, tailoring treatment plans, advancing precision medicine, providing virtual assistant support, and enhancing drug discovery. For SDG7, AI aids in energy consumption forecasting, demand response management, fault detection, and smart grid management. It also enhances the use of renewable energy, improves energy infrastructure reliability, and enables predictive planning for energy resources. In SDG9, AI-powered initiatives promote the development of smart infrastructure, intelligent transportation systems, supply chain management, automated manufacturing processes, and automated quality control. This focused examination highlights the potential impact of AI and big data in these domains, presenting the challenges and future developments required to address key global challenges in health, energy, and infrastructure development.

A. CONTRIBUTIONS OF BIG DATA AND AI TO TOP SDGs (RQ2)

Figure 6 illustrates the important role of AI and big data for SDG3 to strengthen healthcare by enabling more accurate diagnoses, effective treatments, and personalized care. Medical researchers have analyzed extensive datasets via advanced algorithms to increase disease detection and treatment accuracy. For example, the "deep learning in medical data analysis" segment underscores the use of sophisticated AI algorithms for dermatological diagnostics and breast cancer diagnosis, directly contributing to SDG3.8, which emphasizes universal healthcare coverage, including access to quality essential healthcare services. The "AI and big data for Disease Prediction" branch highlights predictive models for chronic kidney disease and heart disease, highlighting how AI-driven insights can lead to early diagnosis and better disease management, thus advancing SDG3.4's target of reducing mortality from noncommunicable diseases through prevention and treatment. This integration of AI and big data enhances diagnostic accuracy and fosters personalized patient care, significantly transforming healthcare delivery systems.

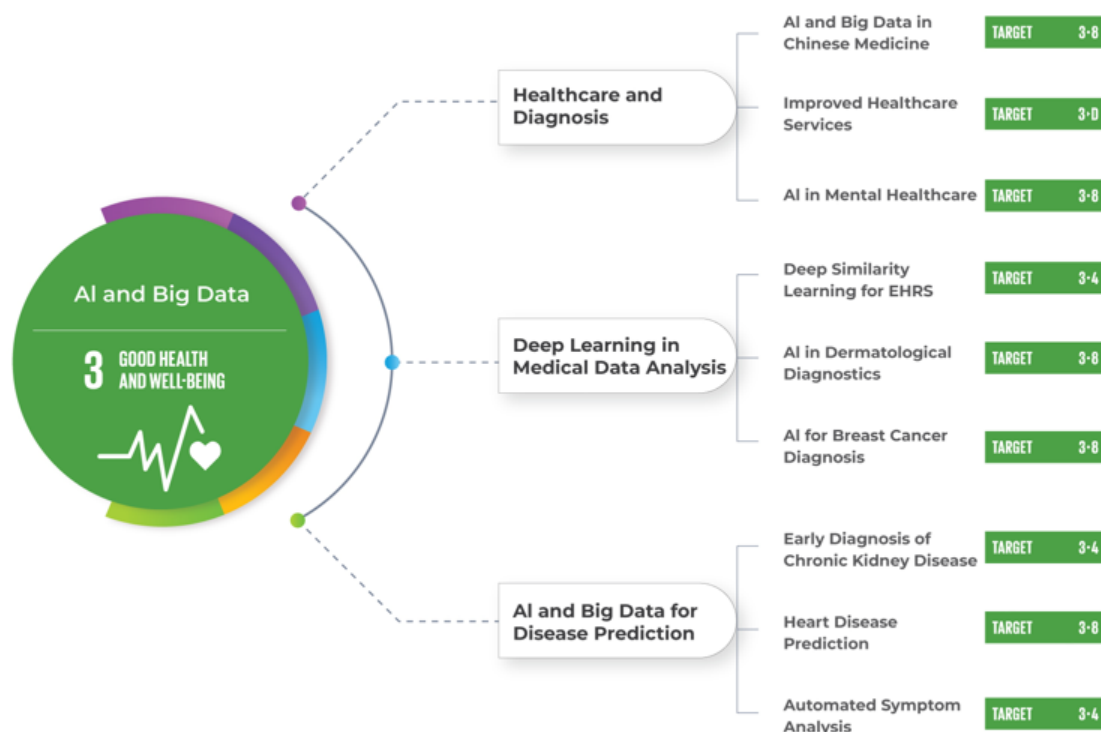


FIGURE 6. Impact of AI and big data on healthcare applications supporting SDG3's goal of ensuring healthy lives and well-being.

Figure 7 shows the transformative potential of AI and big data across various facets of energy management, aiming to advance SDG7, which promotes affordable and clean energy. This approach has diverse applications, from enhancing oil and gas extraction to optimizing energy use in smart buildings and grids. Each branch of the diagram underscores significant innovations such as smart grid reliability and AI-driven models for energy forecasting, highlighting the direct impact on subtargets such as SDG7.3 for improving energy efficiency

and SDG7.1 for increasing renewable energy use. This integration highlights the crucial role of AI and big data in making energy systems more efficient and less environmentally intrusive, highlighting the urgent need for technological advancements to achieve these global goals sustainably.

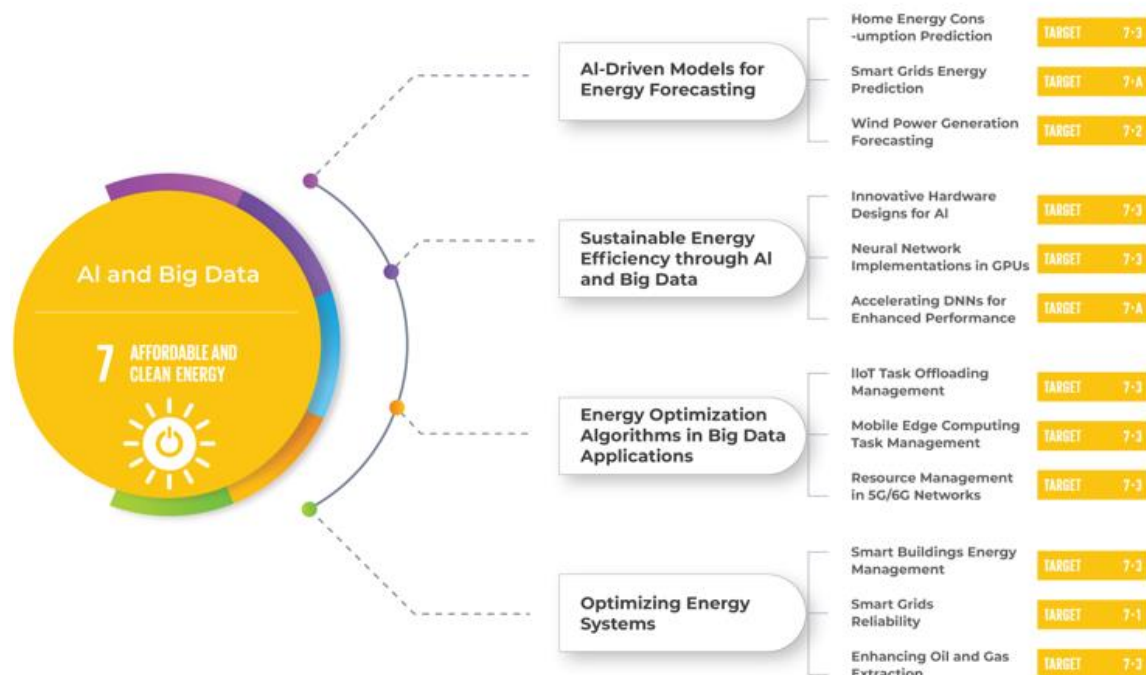


FIGURE 7. Impact of AI and big data in energy management on advancing SDG7 for affordable and clean energy

Figure 8 illustrates the multifaceted impact of AI and big data on industry, innovation, and infrastructure, addressing various components of SDG9. Central themes include optimizing industrial infrastructure through machine health monitoring, network efficiency, and wireless network capabilities, highlighting advancements that bolster infrastructure resilience (SDG9.1) and drive innovation (SDG9.5). The diagram further explores AI-driven big data solutions that enhance cyber-physical

systems and improve traffic network flow prediction, directly fostering innovation within resilient infrastructures. Additionally, intelligent manufacturing in Industry 4.0 is depicted through initiatives such as traffic congestion analysis and smart manufacturing with the IoT, which aligns with SDG9.4 by promoting sustainable industrialization. The segment on predictive maintenance highlights machine learning's role in defect detection and quality control, optimizing industrial processes, and enhancing sustainability.

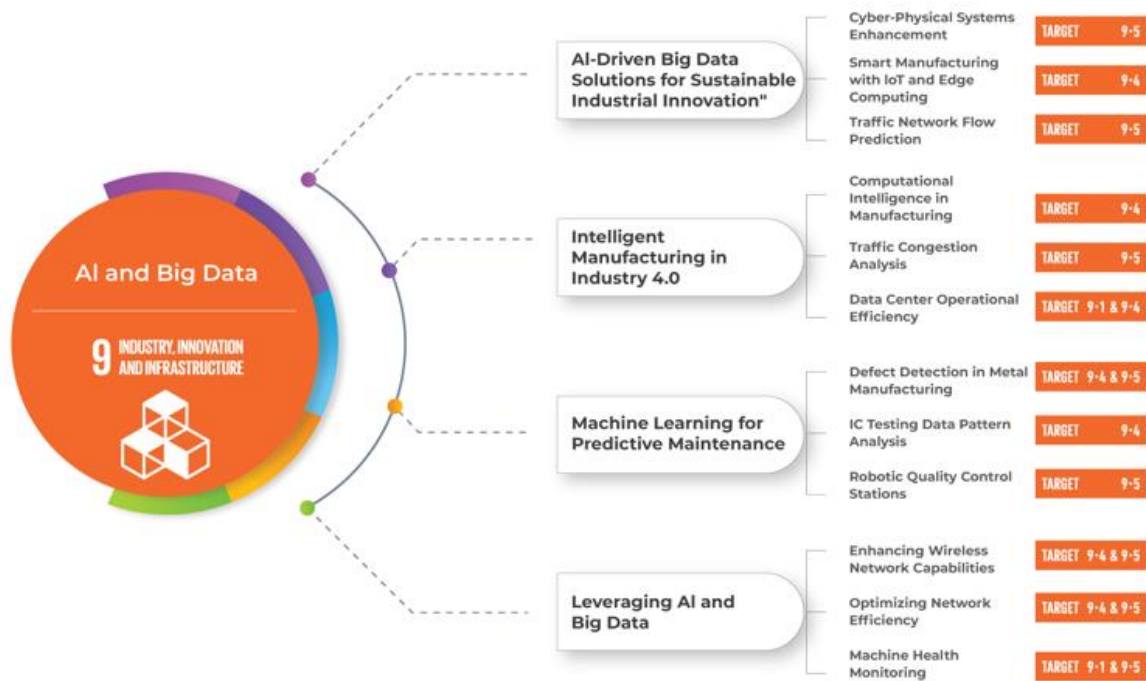


FIGURE 8. Impact of AI and big data in energy management on advancing SDG7 for affordable and clean energy

B. EMERGING TRENDS OF BIG DATA AND AI FOR ADVANCING TOP SDGs (RQ4)

Figure 9 presents a comprehensive framework showcasing the integration of AI and big data algorithms in addressing SDGs 3, 7, and 9. SDG3 has varying data types, such as clinical notes, patient records, medical images such as X-rays and MRIs, wearable sensor data, and genomic data. The primary challenges in this sector include high dimensionality, privacy concerns due to the sensitive nature of health data, heterogeneity from diverse data sources, and incomplete datasets. Moreover, the data might be unstructured, inadequate, or poor quality. Currently, algorithms such as federated learning, capsule networks, and deep learning techniques are utilized to handle image analysis and maintain data privacy while analyzing distributed data sources. Transformers support medical document analysis by finding insights from many clinical notes, research papers, and patient records. Table 1 presents recommended applications of AI techniques, highlighting their strengths and weaknesses and demonstrating their effectiveness in advancing SDGs 3, 7, and 9.

AI-driven predictive genomics and neurosymbolic AI could improve personalized medicine and enhance the interpretability of AI-driven health assessments. Neurosymbolic AI integrates symbolic reasoning with neural networks, offering improved explainability without sacrificing performance, which is crucial in healthcare [104]. Generative AI (GAI) models offer vast potential in healthcare, spanning clinical decision support, personalized treatments, precision

medicine, rehabilitation, medical image translation, lab result translation, drug discovery, automated electronic medical records, medical research, and patient education [105][106]. Different types of GAI algorithms, such as generative adversarial networks for generating synthetic data closely resembling real-world medical images, recurrent neural networks for processing time series medical data, variational autoencoders for generating new data samples useful in drug discovery and personalized medicine, and transformer models for natural language processing tasks such as summarizing clinical notes, transcribing medical records, and analyzing patient-doctor conversations, can be utilized in the health care field to enhance diagnostic capabilities.

SDG7 uses data from energy consumption metrics, smart meter readings, weather conditions influencing energy production, and operational data from energy grids. The challenges here are primarily the variability inherent in renewable energy sources, the extensive volume of data, the temporal ordering of sensor data, high noise content, and the need for real-time processing capabilities. Quantum neural networks and reinforcement learning are currently employed to increase the efficiency of smart grids and optimize energy distribution. Future approaches could involve advanced quantum algorithms for real-time, efficient grid optimization, rapid complex computations, and AI solutions for self-managed autonomous energy grids. It is important to prioritize the integration of AI technologies in smart grids and renewable energy utilization to increase energy efficiency. The use of quantum-based learning algorithms for real-time grid optimization is highly effective since it improves energy

grid operations and decision-making while ensuring robust security against cyber threats. Additionally, adopting AI-driven autonomous energy grids to respond to demand fluctuations and optimize energy flow dynamically is essential. The implementation of self-configuring energy grids with preventive maintenance to increase fault tolerance and resilience and minimize disruptions with minimal human intervention is also recommended. Large Language Models for power system queries and decision-making can increase grid efficiency. Furthermore, public awareness initiatives for demand response should be launched to empower consumers to participate in energy conservation efforts actively. Additionally, optimizing the design of renewable energy infrastructure through AI-driven simulations and analysis provides the potential of AI technologies in shaping a resilient energy future.

SDG9 uses manufacturing data such as supply chain logs, infrastructure monitoring data, and urban planning data. The complexity in this field is due to the interconnected nature of the data, the high volume of data, the diversity of formats, and the necessity for high accuracy in planning and management. Graph neural networks and neural architecture search are key technologies used today to optimize logistics and supply

chains and automate the design of AI models for industrial processes. Additionally, neural architecture search (NAS) supports industrial applications by automating the design of optimized AI models that enhance manufacturing processes and innovation. Reinforcement learning is employed in infrastructure monitoring to optimize maintenance schedules and resource allocation. In the future, cognitive automation could transform manufacturing processes and AI-driven urban planning systems, enabling more sustainable and efficient urban development.

In smart industrialization, AI algorithms analyze manufacturing data to optimize production processes and enable predictive maintenance. GAI virtual assistants enhance knowledge retrieval by facilitating quick access to relevant data for decision-making. Job sequence planning is optimized through AI algorithms to schedule tasks and minimize resource waste. Moreover, AI-driven status report generation automates reporting processes, providing stakeholders with real-time insights into project progress and performance. Safety awareness is improved through AI-powered systems that analyze sensor data and provide proactive alerts and recommendations to mitigate risks.

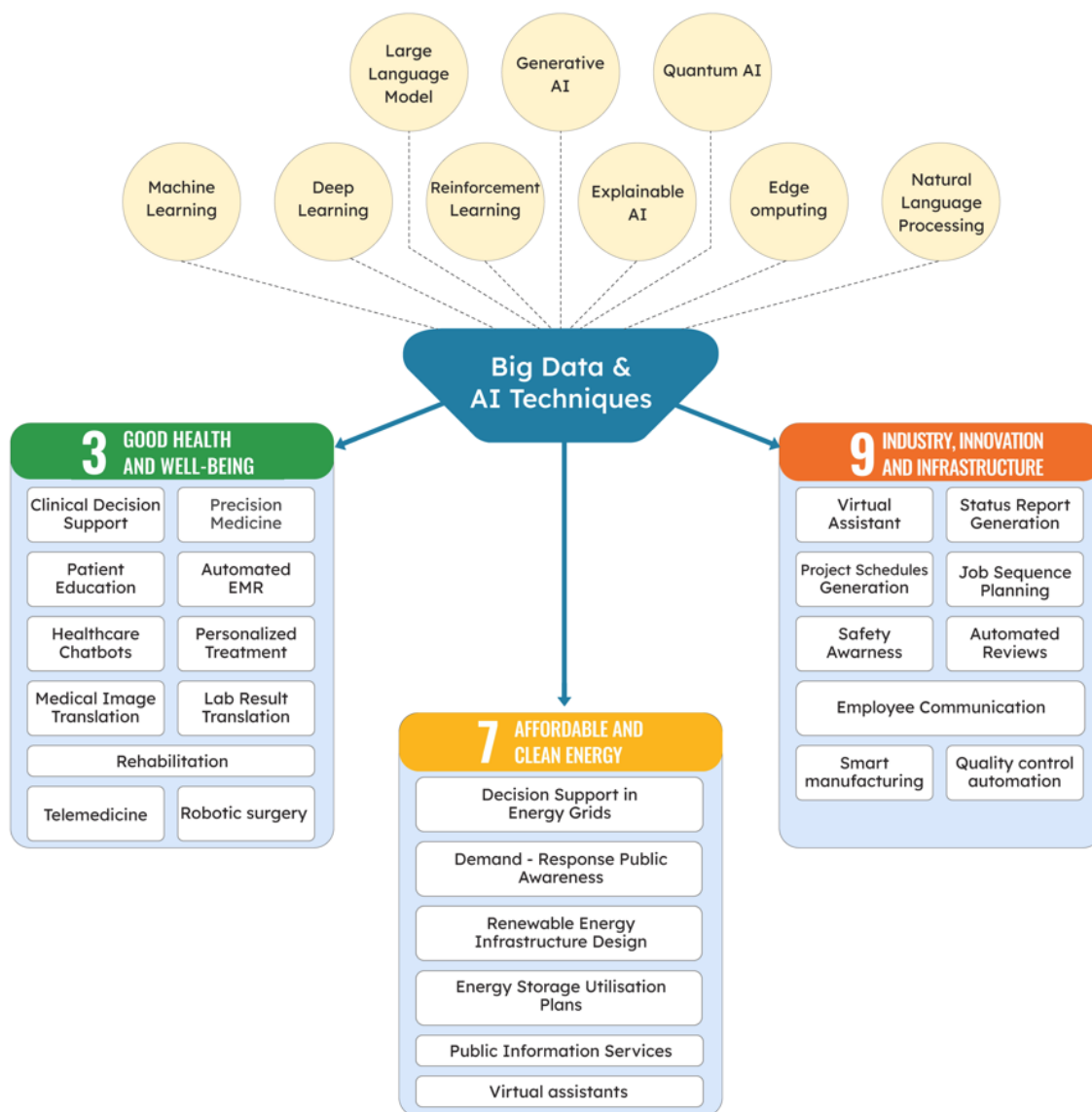


FIGURE 9. Emerging trends of AI and big data algorithms and applications for SDG3, SDG7 and SDG9.

Table 1: Recommended applications of AI techniques, highlighting their strengths and weaknesses, and demonstrating their effectiveness in advancing SDGs 3, 7, and 9

Techniques	SDG	Recommended Applications	Algorithms	Strengths	Weaknesses
Machine Learning	3 GOOD HEALTH AND WELL-BEING [72-74]	▪ Disease tracking ▪ Personalized treatment ▪ Disease Progression ▪ Precision medicine ▪ Drug discovery	▪ Random Forest ▪ Support Vector Machines (SVM) ▪ Gradient boosting models ▪ Regression Models ▪ Time Series Analysis ▪ Clustering algorithms such as k-means ▪ Bayesian Classifier ▪ Ensemble Methods	▪ Identifies patterns in large datasets. ▪ High accuracy in diagnostics ▪ Disease progression analysis ▪ Personalized medicine based on genetic profiles.	▪ Requires large, labeled datasets for training ▪ Risk of overfitting ▪ Ethical issues around data privacy
	7 AFFORDABLE AND CLEAN ENERGY [82-83]	▪ Energy consumption forecasting ▪ Demand response management ▪ Fault Detection	▪ Regression Models ▪ Classification Models such as SVM, k-nn, random forest, logistic regression ▪ Time Series Analysis	▪ Identify patterns and trends in the data ▪ Predicts energy needs ▪ Optimizes resource allocation	▪ Good data quality is required ▪ Data security concerns

		Smart Grid Management	- Clustering Algorithms such as k-means, DBSCAN - Ensemble Methods		
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [93]	- Automated manufacturing processes - Supply chain optimization	- Regression Models - Classification Models such as SVM, k-nn, random forest - Time Series Models - Clustering Algorithms such as k-means - Streaming algorithms	- Enhances operational efficiency - Reduces waste	- Data availability and implementation challenges - Data security concerns
Deep Learning	3 GOOD HEALTH AND WELL-BEING [75-77]	- Medical image analysis - Medical Signal Analysis - Drug discovery	- Convolutional Neural Networks - Recurrent Neural Networks - Autoencoders - Long Short-Term Memory Networks - Attention Mechanisms	- High accuracy in image and signal analysis - Scalable with a large volume of data	- High computational resources required - Requires high-quality, annotated image datasets - Lack of interpretability of results
	7 AFFORDABLE AND CLEAN ENERGY [85-88]	- Smart grid management - Renewable energy forecasting - Remote sensing data analysis - Thermal image analysis of solar panel data	- Convolutional Neural Networks - Recurrent Neural Networks - Long Short-Term Memory Networks - Deep Belief Network - Auto encoders	- Model nonlinear relationships - Handles large data volumes	Difficulty in real-time deployment
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [92,95, 97, 98]	- Industrial automation - Predictive maintenance - Visual inspections in production lines	- Convolutional Neural Networks - Recurrent Neural Networks - Auto encoders - Restricted Boltzmann Machine	- Improves operational efficiency - Reduces downtime	Resource-intensive algorithms
Natural Language Processing	3 GOOD HEALTH AND WELL-BEING [110, 111]	- Electronic Health Record Analysis - Automating clinical documentation - Virtual Assistant support for patient education	- Text Classification Algorithms - Named Entity Recognition - Sentiment Analysis - Topic Modeling - Machine Translation	Extracts insights from unstructured clinical notes and data	Requires large, diverse datasets that cover the medical terminology.
	7 AFFORDABLE AND CLEAN ENERGY [112, 113]	- Virtual assistants to educate consumers - Energy management reports - Customer feedback analysis - Summaries of energy consumption	- Text Summarization - Machine Translation - Topic Modeling - Sentiment Analysis	- Analyze large volumes of text - Extracts effective summaries	- Requires a substantial amount of labeled data for training. - Context sensitivity
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [114, 115]	- Analyzing industry reports - Automating quality control documentation	- Text Summarization - Machine Translation	Enhances information retrieval	Large volumes of domain-specific data required
Reinforcement Learning	3 GOOD HEALTH AND WELL-BEING [116,117]	- Personalized treatment plans - Resource management	- Deep Q-Networks - Policy Gradient Methods - Actor-Critic Algorithms - Proximal Policy Optimization	Adaptive learning with feedback	Requires large data and long training.

	7 AFFORDABLE AND CLEAN ENERGY [118]	▪ Demand-response management ▪ Optimal configurations for maximizing renewable energy resources ▪ Smart grid management	▪ Deep Q-Networks ▪ Policy Gradient Methods ▪ Actor-Critic Algorithms ▪ Proximal Policy Optimization	▪ Can dynamically adjust strategies for fluctuating energy demand response patterns	▪ High resource usage during training ▪ Safety and reliability concerns
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [119,120]	▪ Automate manufacturing processes under dynamic environments ▪ Route planning, inventory management resource allocation in the supply chain	▪ Proximal Policy Optimization ▪ Actor-Critic Algorithms ▪ Policy Gradient Methods	▪ Optimizes resource allocation ▪ Effective in dynamic environments	▪ Requires precise environment modeling
	3 GOOD HEALTH AND WELL-BEING [78]	▪ Health Informatics ▪ Disease Surveillance	▪ Distributed storage and processing ▪ Cluster computing system ▪ Data Warehousing ▪ Cloud Computing	▪ Manage large volumes of data ▪ Real-time data processing ▪ Heterogeneous data integration ▪ Visualize complex healthcare data	▪ Issues with privacy and regulatory compliance
Big Data Analytics	7 AFFORDABLE AND CLEAN ENERGY [91]	▪ Energy market analysis ▪ Demand response analysis	▪ Distributed data processing ▪ Stream processing framework	▪ Facilitates real-time analytics ▪ Integrates diverse data sources	▪ Data privacy and regulatory concerns
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [94, 97]	▪ Analyze manufacturing data ▪ Tracking supply chain	▪ Stream Processing ▪ Predictive Analytics Models	▪ Identifies patterns in the manufacturing process	▪ Dependence on data quality
	3 GOOD HEALTH AND WELL-BEING [118]	▪ Robotic surgery ▪ Robotic rehabilitation	▪ Reinforcement Learning ▪ Motion Planning Algorithms	▪ Enhances precision in surgery ▪ Minimally invasive, improving patient recovery time.	▪ High cost ▪ Ethical concerns
Robotics	7 AFFORDABLE AND CLEAN ENERGY [122]	▪ Maintenance of renewable energy sources	▪ Robotic Process Automation Algorithms	▪ Minimizes downtime in maintenance tasks	▪ Limited operations in unknown environments
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [123]	▪ Automating industrial processes ▪ Maintenance and Inspection ▪ Quality control	▪ Motion Planning Algorithms ▪ Autonomous navigation	▪ Enhances production efficiency ▪ Reduces waste	▪ Vulnerable to security threats
	3 GOOD HEALTH AND WELL-BEING [121]	▪ Tele medicine ▪ Virtual consultations ▪ Patient remote monitoring	▪ Chatbots with NLP ▪ Video-based diagnostics ▪ Predictive analytics models	▪ Expands access to care in remote areas. ▪ Reduces requirements on healthcare infrastructure.	▪ Data security and privacy concerns.
Connected Devices (IoT)	7 AFFORDABLE AND CLEAN ENERGY [122]	▪ Smart meters ▪ Energy management systems	▪ Adaptive data routing ▪ Predictive Maintenance	▪ Improves energy efficiency ▪ Supports demand-side management	▪ Requires robust security measures
	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE [123]	▪ Smart manufacturing and automation ▪ Smart city infrastructure ▪ Urban planning	▪ Edge Computing ▪ Reliable communication ▪ Authentication	▪ Improves operational efficiency	▪ Security issues

C. ROLE OF INTERDISCIPLINARY APPROACHES IN ACHIEVING SDGs (RQ3)

Interdisciplinary approaches that integrate big data and AI enhance the achievement of the SDGs by utilizing expertise from diverse fields. For SDG3, a collaborative effort among healthcare professionals, biologists, pharmacists, public health workers, data scientists, and AI engineers enhances health outcomes. Healthcare professionals, including doctors and nurses, help data scientists identify the most relevant health data for analysis. Biologists and pharmacists provide their knowledge of human biology and drug interactions, which is important for new drug discovery. Public health workers collect data on health trends within communities, which data scientists then analyze to forecast potential outbreaks and identify risk factors. AI engineers design algorithms that can handle large amounts of health data to create predictive models.

In SDG7, engineers design renewable energy systems to ensure a more reliable energy supply. Environmental scientists provide information on the ecological impacts of energy production and provide insights into the sustainability of renewable energy. These insights can help engineers and policymakers create energy solutions that minimize ecological harm. Data scientists and AI engineers analyze real-time data to forecast various factors, such as energy demand and resource utilization. They also provide decision support to improve energy management and efficiency. Similarly, engineers, environmental scientists, data scientists, AI engineers, urban planners, and architects work together to achieve SDG9 by integrating their diverse expertise in building resilient infrastructure and promoting sustainable industrialization.

D. ETHICAL AND SECURITY CHALLENGES IN BIG DATA AND AI FOR SDG IMPLEMENTATION (RQ5)

Data privacy and informed consent are important ethical issues in healthcare when AI is used. These systems collect such personal health information to improve treatments and predict disease conditions. However, many patients do not know how their data might be shared. This lack of understanding can lead to distrust in healthcare systems. Additionally, owing to the sensitive nature of healthcare data, these data can be vulnerable to cyber threats. Wearable sensors often send data over wireless networks, which can be attacked. If the data is not properly encrypted while being transmitted, it may lead to unauthorized access, putting sensitive health information at risk.

In the context of SDG7, data security threats are critical in areas such as smart grids, where interconnected devices depend on real-time data for monitoring and control. Cyberattacks targeting these systems can result in significant disruptions to the power supply and damage to infrastructure. Additionally, in SDG9, applications such as industrial automation rely on data from various sensors and devices to

make real-time decisions. If these data are manipulated or corrupted, it can lead to incorrect decisions, resulting in operational failures or safety hazards.

To address these ethical and security issues, healthcare organizations must use data privacy frameworks. In healthcare systems, patients should be informed about how their data will be used and the risks involved. It is mandatory to invest in cybersecurity measures to protect sensitive information and conduct regular security audits to find vulnerabilities in these systems.

E. IMPLICATIONS FOR PRACTICE (RQ4)

Big data techniques and advanced algorithms have the following implications for practice in healthcare, energy management, manufacturing, urban planning, and interdisciplinary research.

Healthcare: For SDG3, the use of machine learning algorithms such as support vector machines and XGBoost has revolutionized diagnostic processes, enabling earlier and more accurate diagnoses. Additionally, deep learning models can analyze medical images with high accuracy, reducing the time healthcare professionals spend on data processing. NLP and LLMs assist in interpreting clinical notes, benefiting the healthcare team and providing tools for patient education.

The integration of neurosymbolic AI into healthcare practice holds significant potential for improving patient outcomes and enhancing clinical decision-making. By providing more interpretable AI models, healthcare providers can better trust AI-driven decisions, particularly in complex cases such as neurodegenerative diseases and cancer [124]. With the potential for personalized medicine, neurosymbolic AI processes large datasets, such as genetic and imaging data, to provide tailored treatment recommendations while ensuring healthcare providers understand the reasoning behind AI decisions [125]. A significant challenge in neurosymbolic AI is the complexity of combining neural networks with symbolic reasoning. This process involves ensuring seamless interaction between these two AI components, which can be difficult due to the differing nature of learning and reasoning mechanisms [126]. Data interoperability is another major challenge, as healthcare data comes in various formats, including imaging data, clinical notes, and genetic information.

This integration has led to improved patient outcomes and more personalized treatment plans. However, integrating these technologies raises critical questions about the transparency and fairness of the algorithms used, especially given their potential to impact life-altering decisions. The ethical implications and the need for robust, transparent frameworks to integrate these technologies into healthcare systems present vital areas for ongoing research. GAI could enhance healthcare by generating synthetic data and simulations for training, improving diagnostics, and accelerating drug development.

Energy Management: For SDG7, AI-driven smart grids and renewable energy sources are recommended to utilize machine learning and deep learning algorithms for data trend analysis and predictions. Adaptive decision-making is achieved through feedback-based techniques such as reinforcement learning. AI systems can be utilized to predict equipment failures, schedule maintenance, and forecast energy demands while dynamically managing loads. Additionally, these systems can be used to improve smart metering, enabling consumers to monitor their energy usage in real-time and developing energy-saving behaviors. However, these systems rely heavily on the quality and integrity of data, making the scalability and reliability of these algorithms a major area for future research. Addressing the limitations related to data quality and algorithm performance in diverse operational environments is crucial to advancing sustainable energy management. The GAI's predictive models could optimize the integration of renewable energies, enhancing grid efficiency and sustainability.

Manufacturing: The integration of AI in Industry 4.0 offers important practical implications across a range of applications. AI-driven tools can also be employed for project scheduling and resource optimization. Virtual assistants enhance collaboration among team members, leading to improved productivity. By automating the generation of status reports, AI provides real-time updates to make quick decisions. Additionally, safety awareness can be enhanced through predictive analytics of hazards before they occur. Automation in quality control processes helps to maintain product standards and reduce defects. AI enables predictive maintenance and optimizes supply chains, leading to significant efficiency improvements. However, the scalability of these AI solutions is a major concern, as is their ability to maintain performance across different scales and setups. Future research should focus on developing robust AI solutions that can be scaled without loss of efficiency or accuracy, a critical step toward fully realizing Industry 4.0.

Urban Planning: For SDG9, applying GIS and machine learning algorithms in urban planning enhances the ability to predict and plan effectively for urban growth. However, the use of these technologies introduces significant challenges related to data privacy and the ethical management of surveillance data. Future research must address the balance between leveraging AI for public goods and protecting individual privacy rights, ensuring that urban planning practices remain innovative and ethical. The GAI could simulate urban development and traffic systems, aiding planners in creating more efficient, sustainable cities.

Our research has certain limitations. When selecting publications via the PRISMA framework, there is a risk of inherent bias stemming from the specific database chosen.

With respect to BERT topic modeling, we recognize its ability to process extensive datasets and identify central themes. By creating cohesive topic representations through semantic word and phrase similarities, topic modeling reduces human bias in topic classification, thereby enhancing the objectivity and dependability of the outcomes [127]. However, the efficacy of topic modeling algorithms can be influenced by the quality of the input data and the underlying assumptions of their design. Furthermore, mapping publications to SDGs is a complex task, and our analysis acknowledges the shortcomings of relying solely on a particular SDG mapping approach [128]. For future research, comparing our findings with results from various SDG mapping initiatives could offer a more comprehensive perspective and increase the reliability of future studies.

CONCLUSION AND FUTURE DIRECTIONS

The application of big data and AI, specifically through sophisticated algorithms, is revolutionizing practices across multiple sectors that impact SDG3, SDG7, and SDG9. First, our findings demonstrate that AI-powered systems for disease prediction and diagnosis markedly increase the quality and accessibility of healthcare, thereby contributing to the realization of SDG3 [70][75]. Moreover, the results indicated that AI-powered smart grids and energy management systems are driving the development of more adaptive and efficient energy systems, thereby contributing to the realization of SDG7 [69][85]. Furthermore, the application of AI in Industry 4.0, such as predictive maintenance and supply chain optimization, contributes to SDG9 [100][102]. In conclusion, applying big data and AI in interdisciplinary research and sustainable development presents significant ethical and privacy challenges, necessitating a balance between harnessing technology for public benefit and protecting individual privacy. Technology facilitates a deeper understanding and ability to act on complex datasets, improving efficiency, personalization, and predictive capabilities in healthcare, urban planning, and energy management. Importantly, while big data and AI technology can potentially transform many aspects of our lives, there are also associated risks. These risks include ethical implications, data privacy, and the potential for bias in algorithmic decisions.

To effectively harness the potential of big data and AI for advancing the SDGs, this study highlights key recommendations. Additionally, addressing ethical concerns around data privacy and security is critical, especially in sectors that handle sensitive information. The development of robust privacy frameworks and transparent AI algorithms is essential for ensuring public trust. Finally, interdisciplinary collaboration among data scientists, engineers, healthcare professionals, and policymakers is crucial for developing scalable AI-driven strategies to address the multifaceted

challenges posed by the SDGs and foster sustainable development.

Future Research Directions: Several future directions are provided as follows. First, future research can continue to develop transparent and fair AI algorithms and integrate these technologies into a robust framework within healthcare systems. For example, [85] proposed the use of machine learning and data fusion techniques to analyze manufacturing datasets for detecting quality events and enhancing process monitoring. The emerging field of AI-driven predictive genomics opens exciting prospects for precision medicine. Ensuring that AI recommendations are explainable and interpretable requires transparent models and ethical frameworks. AI-enabled simulations could accelerate drug discovery and enhance clinical decision-making.

Second, future research can enhance the scalability, resource efficiency, and reliability of AI models, particularly in the healthcare and energy sectors, where technologies such as neurosymbolic AI and quantum neural networks show promise but remain underexplored. Integrating quantum learning for cybersecurity, self-configuring architectures, and AI for preventive maintenance is pivotal given current threats. The use of distributed big data analytics and imbalanced data classification to improve the reliability and efficiency of smart grids [82]. AI can improve grid reliability and resilience, especially with real-time optimization. Optimizing renewable energy infrastructure through AI can also mitigate environmental impacts.

Third, the advancement of Industry 4.0 through the application of big data and AI is suggested. [86] discussed the potential for leveraging big data and AI to optimize the intelligent automation and operational management of large-scale enterprise cloud data centers. Future research could further explore the development of scalable AI solutions that can be adapted to different industrial applications and maintain efficiency and accuracy—coordinated demand forecasting in supply chains and deployed logistics automation tools to streamline industrialization processes.

It is also important to balance leveraging AI to benefit the public and protect individual privacy rights. [116] explored how the Internet of Things and edge computing drove the development of smart manufacturing systems while emphasizing the challenges in privacy protection. Further studies can be conducted on a mechanism that promotes innovation and provides sound ethical and privacy protection.

REFERENCES

- [1] Y. Xu *et al.*, “Artificial Intelligence: a Powerful Paradigm for Scientific Research,” *The Innovation*, vol. 2, no. 4, Oct. 2021, doi: <https://doi.org/10.1016/j.xinn.2021.100179>.
- [2] R. Nishant, M. Kennedy, and J. Corbett, “Artificial intelligence for sustainability: Challenges, opportunities, and a research agenda,” *International Journal of Information Management*, vol. 53, no. 53, p. 102104, Aug. 2020, doi: <https://doi.org/10.1016/j.ijinfomgt.2020.102104>.
- [3] M. Lowe, R. Qin, and X. Mao, “A Review on Machine Learning, Artificial Intelligence, and Smart Technology in Water Treatment and Monitoring,” *Water*, vol. 14, no. 9, p. 1384, Jan. 2022, doi: <https://doi.org/10.3390/w14091384>.
- [4] D. Rangel-Martinez, K. D. P. Nigam, and L. A. Ricardez-Sandoval, “Machine learning on sustainable energy: A review and outlook on renewable energy systems, catalysis, smart grid and energy storage,” *Chemical Engineering Research and Design*, vol. 174, pp. 414–441, Oct. 2021, doi: <https://doi.org/10.1016/j.cherd.2021.08.013>.
- [5] M. Wählisch, “Big Data, New Technologies, and Sustainable Peace: Challenges and Opportunities for the UN,” *Journal of Peacebuilding & Development*, vol. 15, no. 1, p. 154231661986898, Aug. 2019, doi: <https://doi.org/10.1177/1542316619868984>.
- [6] M. Naeem *et al.*, “Trends and Future Perspective Challenges in Big Data,” *Advances in Intelligent Data Analysis and Applications*, vol. 253, pp. 309–325, Nov. 2021, Available: https://link.springer.com/chapter/10.1007/978-16-5036-9_30
- [7] A. Adadi, “A Survey on Data-efficient Algorithms in Big Data Era,” *Journal of Big Data*, vol. 8, no. 1, Jan. 2021, doi: <https://doi.org/10.1186/s40537-021-00419-9>.
- [8] A. Holzinger, K. Keiblinger, P. Holub, K. Zatloukal, and H. Müller, “AI for life: Trends in artificial intelligence for biotechnology,” *New Biotechnology*, vol. 74, pp. 16–24, May 2023, doi: <https://doi.org/10.1016/j.nbt.2023.02.001>.
- [9] H. Luan *et al.*, “Challenges and Future Directions of Big Data and Artificial Intelligence in Education,” *Frontiers in Psychology*, vol. 11, no. 11, Oct. 2020, doi: <https://doi.org/10.3389/fpsyg.2020.580820>.
- [10] S. Yu and J. Ma, “Deep Learning for Geophysics: Current and Future Trends,” *Reviews of Geophysics*, vol. 59, no. 3, Jul. 2021, doi: <https://doi.org/10.1029/2021rg000742>.
- [11] C. Caudai *et al.*, “AI applications in functional genomics,” *Computational and Structural Biotechnology Journal*, vol. 19, pp. 5762–5790, 2021, doi: <https://doi.org/10.1016/j.csbj.2021.10.009>.
- [12] L. E. Mansuri and D. A. Patel, “Artificial intelligence-based automatic visual inspection system for built heritage,” *Smart and Sustainable Built Environment*, vol. ahead-of-print, no. ahead-of-print, Feb. 2021, doi: <https://doi.org/10.1108/sasbe-09-2020-0139>.
- [13] Ó. Álvarez-Machancoses, E. J. DeAndrés Galiana, A. Cernea, J. Fernández de la Viña, and J. L. Fernández-Martínez, “On the Role of Artificial Intelligence in Genomics to Enhance Precision Medicine,” *Pharmacogenomics and Personalized Medicine*, vol. 13, pp. 105–119, Mar. 2020, doi: <https://doi.org/10.2147/PGPM.S205082>.
- [14] G. Furano *et al.*, “Towards the Use of Artificial Intelligence on the Edge in Space Systems: Challenges and Opportunities,” *IEEE Aerospace and Electronic Systems Magazine*, vol. 35, no. 12, pp. 44–56, Dec. 2020, doi: <https://doi.org/10.1109/MAES.2020.3008468>.
- [15] M. R. Dobbelaere, P. P. Plehiers, R. Van de Vijver, C. V. Stevens, and K. M. Van Geem, “Machine Learning in Chemical Engineering: Strengths, Weaknesses, Opportunities, and Threats,” *Engineering*, vol. 7, no. 9, Jul. 2021, doi: <https://doi.org/10.1016/j.eng.2021.03.019>.
- [16] L. Sun, Z. Shang, Y. Xia, S. Bhowmick, and S. Nagarajaiah, “Review of Bridge Structural Health Monitoring Aided by Big Data and Artificial Intelligence: From Condition Assessment to Damage Detection,” *Journal of Structural Engineering*, vol. 146, no. 5, p. 04020073, May 2020, doi: [https://doi.org/10.1061/\(asce\)st.1943-541x.0002535](https://doi.org/10.1061/(asce)st.1943-541x.0002535).
- [17] Zoran Babović *et al.*, “Research in computing-intensive simulations for nature-oriented civil-engineering and related scientific fields, using machine learning and big data: an overview of open problems,” *Journal of Big Data*, vol. 10, no. 1, May 2023, doi: <https://doi.org/10.1186/s40537-023-00731-6>.
- [18] Miroslav Kosanic and V. Milutinovic, “A Survey on Mathematical Aspects of Machine Learning in GeoPhysics: The Cases of Weather Forecast, Wind Energy, Wave Energy, Oil and Gas Exploration,” *arXiv (Cornell University)*, Jun. 2021, doi: <https://doi.org/10.1109/meco52532.2021.9460245>.
- [19] J. T. Hancock, T. M. Khoshgoftaar, and J. M. Johnson, “Evaluating classifier performance with highly imbalanced Big Data,” *Journal of Big Data*, vol. 10, no. 1, Apr. 2023, doi: <https://doi.org/10.1186/s40537-023-00724-5>.

- [20] J. Hancock, H. Wang, T. M. Khoshgoftaar, and Q. Liang, "Data reduction techniques for highly imbalanced medicare Big Data," *Journal of Big Data*, vol. 11, no. 1, Jan. 2024, doi: <https://doi.org/10.1186/s40537-023-00869-3>.
- [21] F. F. Costa, "Big data in biomedicine," *Drug Discovery Today*, vol. 19, no. 4, pp. 433–440, Apr. 2014, doi: <https://doi.org/10.1016/j.drudis.2013.10.012>.
- [22] Zhao, L., Ciallrella, H. L., Aleksunes, L. M., & Zhu, H. "Advancing computer-aided drug discovery (CADD) by big data and data-driven machine learning modeling," *Drug Discovery Today*, vol. 25, no. 9, pp. 1624–1638, Sep. 2020, doi: <https://doi.org/10.1016/j.drudis.2020.07.005>.
- [23] Z. Engin et al., "Data-driven urban management: Mapping the landscape," *Journal of Urban Management*, vol. 9, no. 2, pp. 140–150, Jun. 2020, doi: <https://doi.org/10.1016/j.jum.2019.12.001>.
- [24] J. Kandt and M. Batty, "Smart cities, big data and urban policy: Towards urban analytics for the long run," *Cities*, vol. 109, no. 102992, p. 102992, Nov. 2020, doi: <https://doi.org/10.1016/j.cities.2020.102992>.
- [25] D. Mhlanga, "Artificial Intelligence in the Industry 4.0, and Its Impact on Poverty, Innovation, Infrastructure Development, and the Sustainable Development Goals: Lessons from Emerging Economies?," *Sustainability*, vol. 13, no. 11, p. 5788, May 2021, doi: <https://doi.org/10.3390/su13115788>.
- [26] R. Kusters et al., "Interdisciplinary Research in Artificial Intelligence: Challenges and Opportunities," *Frontiers in Big Data*, vol. 3, Nov. 2020, doi: <https://doi.org/10.3389/fdata.2020.577974>.
- [27] J. Bajorath, "Artificial intelligence in interdisciplinary life science and drug discovery research," *Future Science OA*, vol. 8, no. 4, Apr. 2022, doi: <https://doi.org/10.2144/fsoa-2022-0010>.
- [28] N. M. Safdar, J. D. Banja, and C. C. Meltzer, "Ethical Considerations in Artificial Intelligence," *European Journal of Radiology*, vol. 122, no. 1, p. 108768, Jan. 2020, Available: <https://www.sciencedirect.com/science/article/pii/S0720048X19304188>.
- [29] B. Murdoch, "Privacy and Artificial Intelligence: Challenges for Protecting Health Information in a New Era," *BMC Medical Ethics*, vol. 22, no. 1, Sep. 2021, doi: <https://doi.org/10.1186/s12910-021-00687-3>.
- [30] J. M. Johnson and T. M. Khoshgoftaar, "Data-Centric AI for Healthcare Fraud Detection," *SN Computer Science*, vol. 4, no. 4, May 2023, doi: <https://doi.org/10.1007/s42979-023-01809-x>.
- [31] D. Liu, H. Kong, X. Luo, W. Liu, and R. Subramaniam, "Bringing AI to edge: From deep learning's perspective," *Neurocomputing*, vol. 485, pp. 297–320, May 2022, doi: <https://doi.org/10.1016/j.neucom.2021.04.141>.
- [32] S. Weixing, L.-F. Li, F. Liu, M. He, and L. Lin, "AI on the edge: a comprehensive review," *Artificial Intelligence Review*, vol. 55, no. 8, pp. 6125–6183, Mar. 2022, doi: <https://doi.org/10.1007/s10462-022-10141-4>.
- [33] R. Walshe, A. Koene, S. Baumann, M. Panella, L. Maglaras, and F. Medeiros, "Artificial Intelligence as Enabler for Sustainable Development," *2021 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)*, Jun. 2021, doi: <https://doi.org/10.1109/ice/itmc52061.2021.9570215>.
- [34] Z. Chen, M. Wu, A. Chan, X. Li, and Y.-S. Ong, "Survey on AI Sustainability: Emerging Trends on Learning Algorithms and Research Challenges [Review Article]," *IEEE Computational Intelligence Magazine*, vol. 18, no. 2, pp. 60–77, May 2023, doi: <https://doi.org/10.1109/mci.2023.3245733>.
- [35] F. Rohde, M. Gossen, J. Wagner, and T. Santarius, "Sustainability challenges of Artificial Intelligence and Policy Implications," *Ökologisches Wirtschaften - Fachzeitschrift*, vol. 36, no. 01, pp. 36–40, Feb. 2021, doi: <https://doi.org/10.14512/oewo360136>.
- [36] H. Guo, H. Hackmann, and K. Gong, "Big data in support of the Sustainable Development Goals: a celebration of the establishment of the International Research Center of Big Data for Sustainable Development Goals (CBAS)," *Big Earth Data*, vol. 5, no. 3, pp. 259–262, Jul. 2021, doi: <https://doi.org/10.1080/20964471.2021.1962621>.
- [37] Rishikesh Bamdale, S. Shelar, and V. Khandekar, "How to tackle Climate Change using Artificial Intelligence," Jul. 2021, doi: <https://doi.org/10.1109/iccncnt51525.2021.9579674>.
- [38] D. Si, "A Framework to analyze the Impacts of AI with the Sustainable Development Goals," *Highlights in Science, Engineering and Technology*, vol. 17, pp. 313–323, Nov. 2022, doi: <https://doi.org/10.54097/hset.v17i.2621>.
- [39] M.-L. How, S.-M. Cheah, Y.-J. Chan, A. C. Khor, and E. M. P. Say, "Artificial Intelligence-Enhanced Decision Support for Informing Global Sustainable Development: A Human-Centric AI-Thinking Approach," *Information*, vol. 11, no. 1, p. 39, Jan. 2020, doi: <https://doi.org/10.3390/info11010039>.
- [40] L. Fan, J. Li, and X. P. Zhang, "Machine learning load prediction methods for home energy management systems based on human behavior patterns recognition," *CSEE Journal of Power and Energy Systems*, 2020, doi: <https://doi.org/10.17775/cseejpes.2018.01130>.
- [41] R. Raman, hiran lathabhai, Сантану Мандал, C. Kumar, and Prema Nedungadi, "Contribution of Business Research to Sustainable Development Goals: Bibliometrics and Science Mapping Analysis," *Sustainability*, vol. 15, no. 17, pp. 12982–12982, Aug. 2023, doi: <https://doi.org/10.3390/su151712982>.
- [42] R. Raman et al., "Mapping sustainability reporting research with the UN's sustainable development goal," *Heliyon*, vol. 9, no. 8, pp. e18510–e18510, Aug. 2023, doi: <https://doi.org/10.1016/j.heliyon.2023.e18510>.
- [43] V. K. Singh, P. Singh, M. Karmakar, J. Leta, and P. Mayr, "The journal coverage of Web of Science, Scopus and Dimensions: A comparative analysis," *Scientometrics*, vol. 126, no. 6, pp. 5113–5142, Mar. 2021, doi: <https://doi.org/10.1007/s11192-021-03948-5>.
- [44] ANZSCO, *Australian and New Zealand Standard Classification of Occupations (ANZSCO)*, Version 1.2, Australian Bureau of Statistics, Canberra, 2013.
- [45] R. Raman et al., "Fake news research trends, linkages to generative artificial intelligence and sustainable development goals," *Heliyon*, vol. 10, no. 3, p. e24727, Feb. 2024, doi: <https://doi.org/10.1016/j.heliyon.2024.e24727>.
- [46] R. Raman, D. Pattnaik, C. Kumar, and P. Nedungadi, "Advancing sustainable energy systems: A decade of SETA research contribution to sustainable development goals," *Sustainable Energy Technologies and Assessments*, vol. 71, p. 103978, Nov. 2024, doi: <https://doi.org/10.1016/j.seta.2024.103978>.
- [47] R. Raman et al., "Mapping research in the Journal of Innovation & Knowledge to sustainable development goals," *Journal of Innovation & Knowledge*, vol. 9, no. 3, p. 100538, Jul. 2024, doi: <https://doi.org/10.1016/j.jik.2024.100538>.
- [48] N. J. van Eck and L. Waltman, "Software survey: VOSviewer, a computer program for bibliometric mapping," *Scientometrics*, vol. 84, no. 2, pp. 523–538, Dec. 2010, doi: <https://doi.org/10.1007/s11192-009-0146-3>.
- [49] R. Egger and J. Yu, "A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts," *Frontiers in Sociology*, vol. 7, May 2022, doi: <https://doi.org/10.3389/fsoc.2022.886498>.
- [50] Maarten Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," *arXiv (Cornell University)*, Mar. 2022, doi: <https://doi.org/10.48550/arxiv.2203.05794>.
- [51] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *arXiv.org*, May 24, 2019, <https://arxiv.org/abs/1810.04805#>.
- [52] Andry Alamsyah and Nadhif Diterian Girawan, "Improving Clothing Product Quality and Reducing Waste Based on Consumer Review Using RoBERTa and BERTopic Language Model," *Big data and cognitive computing*, vol. 7, no. 4, pp. 168–168, Oct. 2023, doi: <https://doi.org/10.3390/bdcc7040168>.
- [53] J. Yi, Yun Kyung Oh, and J.-M. Kim, "Unveiling the drivers of satisfaction in mobile trading: Contextual mining of retail investor experience through BERTopic and generative AI," *Journal of Retailing and Consumer Services*, vol. 82, pp. 104066–104066, Jan. 2025, doi: <https://doi.org/10.1016/j.jretconser.2024.104066>.
- [54] Yun Kyung Oh, J. Yi, and J.-D. Kim, "What enhances or worsens the user-generated metaverse experience? An application of BERTopic to Roblox user eWOM," *Internet Research*, Dec. 2023, doi: <https://doi.org/10.1108/intr-03-2022-0178>.

- [55] K. Kim, D. F. Kogler, and Sira Maliphol, "Identifying interdisciplinary emergence in the science of science: combination of network analysis and BERTopic," *Humanities and Social Sciences Communications*, vol. 11, no. 1, May 2024, doi: <https://doi.org/10.1057/s41599-024-03044-y>.
- [56] Khodeir, N., & Elghannam, F. (2024). Efficient topic identification for urgent MOOC Forum posts using BERTopic and traditional topic modeling techniques. *Education and Information Technologies*, 1-27.
- [57] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction.
- [58] Lyutov, A., Uygun, Y., & Hütt, M. T. (2024). Machine learning misclassification networks reveal a citation advantage of interdisciplinary publications only in high-impact journals. *Scientific Reports*, 14(1), 21906.
- [59] Capra L (2024) A computational linguistic approach to study border theory at scale. *ACM Trans Comput-Hum Interaction* 37(4):1–23
- [60] G. Castellanos, C. Jing, J. Chen, and H. Chen, "Identifying interdisciplinary topics and their evolution based on BERTopic," *Scientometrics*, Jul. 2023, doi: <https://doi.org/10.1007/s11192-023-04776-5>.
- [61] V. O. K. Li, J. C. K. Lam, and J. Cui, "AI for Social Good: AI and Big Data Approaches for Environmental Decision-Making," *Environmental Science & Policy*, vol. 125, pp. 241–246, Nov. 2021, doi: <https://doi.org/10.1016/j.envsci.2021.09.001>.
- [62] R. Lokers, R. Knapen, S. Janssen, Y. van Randen, and J. Jansen, "Analysis of Big Data technologies for use in agro-environmental science," *Environmental Modelling & Software*, vol. 84, pp. 494–504, Oct. 2016, doi: <https://doi.org/10.1016/j.envsoft.2016.07.017>.
- [63] S. E. Bibri, A. Alexandre, A. Sharifi, and J. Krogstie, "Environmentally sustainable smart cities and their converging AI, IoT, and big data technologies and solutions: an integrated approach to an extensive literature review," *Energy Informatics*, vol. 6, no. 1, Apr. 2023, doi: <https://doi.org/10.1186/s42162-023-00259-2>.
- [64] D. Kamrowska-Zaluska, "Impact of AI-Based Tools and Urban Big Data Analytics on the Design and Planning of Cities," *Land*, vol. 10, no. 11, p. 1209, Nov. 2021, doi: <https://doi.org/10.3390/land10111209>.
- [65] M. Maisonnobe, "The future of urban models in the Big Data and AI era: a bibliometric analysis (2000–2019)," *AI & SOCIETY*, Mar. 2021, doi: <https://doi.org/10.1007/s00146-021-01166-4>.
- [66] L. H. Kaack, P. L. Danti, E. Strubell, G. Kamiya, F. Creutzig, and D. Rolnick, "Aligning artificial intelligence with climate change mitigation," *Nature Climate Change*, vol. 12, no. 6, pp. 518–527, Jun. 2022, doi: <https://doi.org/10.1038/s41558-022-01377-7>.
- [67] K. Zhang and A. B. Aslan, "AI technologies for education: Recent research & future directions," *Computers and Education: Artificial Intelligence*, vol. 2, no. 100025, p. 100025, Jun. 2021, doi: <https://doi.org/10.1016/j.caeai.2021.100025>.
- [68] K. G. Srinivasa, M. Kurni, and K. Saritha, "Harnessing the Power of AI to Education," *Springer Texts in Education*, pp. 311–342, 2022.
- [69] T. Ahmad *et al.*, "Artificial Intelligence in Sustainable Energy industry: Status Quo, Challenges and Opportunities," *Journal of Cleaner Production*, vol. 289, no. 289, p. 125834, Mar. 2021, doi: <https://doi.org/10.1016/j.jclepro.2021.125834>.
- [70] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in health and medicine," *Nature Medicine*, vol. 28, no. 1, pp. 31–38, Jan. 2022, doi: <https://doi.org/10.1038/s41591-021-01614-0>.
- [71] S. Qazi, B. A. Khawaja, and Q. U. Farooq, "IoT-Equipped and AI-Enabled Next Generation Smart Agriculture: A Critical Review, Current Challenges and Future Trends," *IEEE Access*, vol. 10, pp. 21219–21235, 2022, doi: <https://doi.org/10.1109/access.2022.3152544>.
- [72] M. A. Abdel-Fattah, N. A. Othman, and N. Goher, "Predicting Chronic Kidney Disease Using Hybrid Machine Learning Based on Apache Spark," *Computational Intelligence and Neuroscience*, vol. 2022, pp. 1–12, Feb. 2022, doi: <https://doi.org/10.1155/2022/9898831>.
- [73] Saad Almutairi, S. Manimurugan, B.-G. Kim, Majed Aborokbah, and C. Narmatha, "Breast cancer classification using Deep Q Learning (DQL) and gorilla troops optimization (GTO)," *Applied Soft Computing*, vol. 142, pp. 110292–110292, Jul. 2023, doi: <https://doi.org/10.1016/j.asoc.2023.110292>.
- [74] A. Mittal, L. Dumka, and L. Mohan, "A Comprehensive Review on the Use of Artificial Intelligence in Mental Health Care," Jul. 2023, doi: <https://doi.org/10.1109/iccmt.2023.10308255>.
- [75] Habeeb Omotunde and M. R. Mouhamed, "The Modern Impact of Artificial Intelligence Systems in Healthcare: A Concise Analysis," *Mesopotamian Journal of Artificial Intelligence in Healthcare*, vol. 2023, pp. 66–70, Nov. 2023, doi: <https://doi.org/10.58496/mjaiih/2023/013>.
- [76] V. Gupta, S. Sachdeva, and S. Bhalla, "A Novel Deep Similarity Learning Approach to Electronic Health Records Data," *IEEE Access*, pp. 1–1, 2020, doi: <https://doi.org/10.1109/access.2020.3037710>.
- [77] M. Elbes, S. AlZu'bi, T. Kanan, A. Mughaid, and S. Abushanab, "Big dermatological data service for precise and immediate diagnosis by utilizing pre-trained learning models," *Cluster Computing*, vol. 27, no. 5, pp. 6931–6951, Mar. 2024, doi: <https://doi.org/10.1007/s10586-024-04331-8>.
- [78] M. Ye, H. Zhang, and L. Li, "Research on Data Mining Application of Orthopedic Rehabilitation Information for Smart Medical," *IEEE Access*, vol. 7, pp. 177137–177147, 2019, doi: <https://doi.org/10.1109/access.2019.2957579>.
- [79] N. Ahmad, Z. Ullah, Hyungseo Bobby Ryu, A. Ariza-Montes, and H. Han, "From Corporate Social Responsibility to Employee Well-Being: Navigating the Pathway to Sustainable Healthcare," *Psychology Research and Behavior Management*, vol. Volume 16, no. 16, pp. 1079–1095, Apr. 2023, doi: <https://doi.org/10.2147/prbm.s398586>.
- [80] Sriya Kanamarlapudi, Venkata Santhosh Yakkala, Badisa Gayathri, Krishna Vamsi Nusalma, S. S. Aravindh, and S. Srithar, "Comparison and Analysis of Various Machine Learning Algorithms for Disease Prediction," Feb. 2023, doi: <https://doi.org/10.1109/iccmt.2023.10083509>.
- [81] G. Braswell, "Artificial Intelligence Comes of Age in Oil and Gas," *Journal of Petroleum Technology*, vol. 65, no. 01, pp. 50–57, Jan. 2013, doi: <https://doi.org/10.2118/0113-0050-jpt>.
- [82] H. Wang, M. Xiao, C. Wu, and J. Zhang, "Distributed classification for imbalanced big data in distributed environments," *Wireless Networks*, Feb. 2021, doi: <https://doi.org/10.1007/s11276-021-02552-y>.
- [83] H. Dong, W. Hongkai, W. Xiaohua, H. Haichao, M. Shule, and G. Yang, "Log fusion technology of power information system based on fuzzy reasoning," *2020 International Conference on Virtual Reality and Intelligent Systems (ICVRIS)*, vol. 43, pp. 296–299, Jul. 2020, doi: <https://doi.org/10.1109/icvr.2020.00076>.
- [84] Qu Xiaoyun, Kang Xiaoning, Zhang Chao, Jiang Shuai, and Ma Xiuda, "Short-term prediction of wind power based on deep Long Short-Term Memory," *2016 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC)*, Oct. 2016, doi: <https://doi.org/10.1109/appeec.2016.7779672>.
- [85] Katarina Grolinger, Miriam, and L. Seewald, "Energy Consumption Prediction with Big Data: Balancing Prediction Accuracy and Computational Resources," *International Congress on Big Data*, Jun. 2016, doi: <https://doi.org/10.1109/bigdatacongress.2016.27>.
- [86] M. A. Hanif, R. Hafiz, M. U. Javed, S. Rehman, and M. Shafique, "Energy-Efficient Design of Advanced Machine Learning Hardware," *Springer eBooks*, pp. 647–678, Jan. 2019, doi: https://doi.org/10.1007/978-3-030-04666-8_21.
- [87] Mohd Saqib Akhoun *et al.*, "High performance accelerators for deep neural networks: A review," *Expert Systems*, vol. 39, no. 1, Oct. 2021, doi: <https://doi.org/10.1111/essy.12831>.
- [88] Z. Wang, Shamma Nasrin, R. Islam, A. Haque, and Muhammed, "Emerging memories and their applications in neuromorphic computing," *Elsevier eBooks*, pp. 305–357, Jan. 2023, doi: <https://doi.org/10.1016/b978-0-323-91832-9.00005-1>.
- [89] G. Wu, Z. Xu, H. Zhang, S. Shen, and S. Yu, "Multi-agent DRL for joint completion delay and energy consumption with queuing theory in MEC-based IIoT," vol. 176, pp. 80–94, Jun. 2023, doi: <https://doi.org/10.1016/j.jpdc.2023.02.008>.
- [90] Z. Shi, X. Xie, S. Garg, H. Lu, H. Yang, and Z. Xiong, "Deep Reinforcement Learning Based Big Data Resource Management for 5G/6G Communications," *2021 IEEE Global Communications Conference (GLOBECOM)*, pp. 01–06, Dec. 2021, doi: <https://doi.org/10.1109/globecom46510.2021.9685098>.
- [91] X. Zhao, Y. Zhang, and M. Li, "Task Migration Optimization Algorithm in Mobile Edge Computing," pp. 133–137, Jun. 2023, doi: <https://doi.org/10.1145/3605801.3605827>.
- [92] I. Ul Haq, S. Anwar, and T. Khan, "Machine Vision Based Predictive Maintenance for Machine Health Monitoring: A Comparative

- Analysis,” *IEEE Xplore*, Mar. 01, 2023, <https://ieeexplore.ieee.org/document/10089572/> (accessed Apr. 26, 2023).
- [93] K. M. Baalamurugan and Aanchal Phutela, “A Brief Study of Adaptive Clustering for Self-aware Machine Analytics,” *Disruptive technologies and digital transformations for society 5.0*, pp. 49–69, Jan. 2024, doi: https://doi.org/10.1007/978-981-99-8118-2_3.
- [94] M. G. Kibria, K. Nguyen, G. P. Villardi, O. Zhao, K. Ishizu, and F. Kojima, “Big Data Analytics, Machine Learning, and Artificial Intelligence in Next-Generation Wireless Networks,” *IEEE Access*, vol. 6, pp. 32328–32338, 2018, doi: <https://doi.org/10.1109/access.2018.2837692>.
- [95] Y. Zhang, Y. Zhou, H. Lu, and H. Fujita, “Traffic Network Flow Prediction Using Parallel Training for Deep Convolutional Neural Networks on Spark Cloud,” *IEEE Transactions on Industrial Informatics*, vol. 16, no. 12, pp. 7369–7380, Feb. 2020, doi: <https://doi.org/10.1109/tii.2020.2976053>.
- [96] M. Abbasi, M. Plaza-Hernandez, J. Prieto, and J. M. Corchado, “Security in the Internet of Things Application Layer: Requirements, Threats, and Solutions,” *IEEE Access*, vol. 10, pp. 97197–97216, 2022, doi: <https://doi.org/10.1109/access.2022.3205351>.
- [97] D. Jeong, “Road Damage Detection Using YOLO with Smartphone Images,” *2020 IEEE International Conference on Big Data (Big Data)*, Dec. 2020, doi: <https://doi.org/10.1109/bigdata50022.2020.9377847>.
- [98] F. Sun, A. Dubey, and J. White, “DxNAT — Deep neural networks for explaining non-recurring traffic congestion,” *2017 IEEE International Conference on Big Data (Big Data)*, Dec. 2017, doi: <https://doi.org/10.1109/bigdata.2017.8258162>.
- [99] N. Thillaiarasu, S. Lata Tripathi, and V. Dhinakaran, *Artificial Intelligence for Internet of Things*. Boca Raton: CRC Press, 2022, doi: <https://doi.org/10.1201/9781003335801>.
- [100] G. Feng and A. Li, “Intelligent Automated Operation and Operation Management of Large Enterprise Cloud Data Center Based on Artificial Intelligence,” *Advances in intelligent systems and computing*, pp. 531–538, Jan. 2021, doi: https://doi.org/10.1007/978-981-16-1726-3_65.
- [101] C. A. Escobar, D. Macias, and R. Morales-Menendez, “Process monitoring for quality — A multiple classifier system for highly unbalanced data,” *Heliyon*, vol. 7, no. 10, p. e08123, Oct. 2021, doi: <https://doi.org/10.1016/j.heliyon.2021.e08123>.
- [102] L. Variz, L. Piardi, P. J. Rodrigues, and P. Leita, “Machine Learning Applied to an Intelligent and Adaptive Robotic Inspection Station,” *2019 IEEE 17th International Conference on Industrial Informatics (INDIN)*, Jul. 2019, doi: <https://doi.org/10.1109/indin41052.2019.8972298>.
- [103] B. C. Wu, “IC Test Quality Enhancement by Introducing Machine Learning,” *2019 Joint International Symposium on e-Manufacturing & Design Collaboration (eMDC) & Semiconductor Manufacturing (ISSM)*, vol. 106, pp. 1–4, Sep. 2019, doi: <https://doi.org/10.23919/emdc/issm48219.2019.9052135>.
- [104] Garcez and L. C. Lamb, “Neurosymbolic AI: the 3rd wave,” Mar. 2023, doi: <https://doi.org/10.1007/s10462-023-10448-w>.
- [105] J. Varghese and J. Chapiro, “ChatGPT: The transformative influence of generative AI on science and healthcare,” *Journal of Hepatology*, Aug. 2023, doi: <https://doi.org/10.1016/j.jhep.2023.07.028>.
- [106] P. Zhang and Maged, “Generative AI in Medicine and Healthcare: Promises, Opportunities and Challenges,” *Future Internet*, vol. 15, no. 9, pp. 286–286, Aug. 2023, doi: <https://doi.org/10.3390/fi15090286>.
- [107] E. Hossain et al., “Use of ai/ml-enabled state-of-the-art method in electronic medical records: A systematic review,” *Computers in Biology and Medicine*, vol. 155, p. 106649, Feb. 2023, doi: <https://doi.org/10.1016/j.combiomed.2023.106649>.
- [108] Y. Juhn and H. Liu, “Artificial intelligence approaches using natural language processing to advance EHR-based clinical research,” *Journal of Allergy and Clinical Immunology*, vol. 145, no. 2, pp. 463–469, Feb. 2020, doi: <https://doi.org/10.1016/j.jaci.2019.12.897>.
- [109] J. Aguilar, A. Garces-Jimenez, M. D. R-Moreno, and R. Garcia, “A systematic literature review on the use of artificial intelligence in energy self-management in smart buildings,” *Renewable and Sustainable Energy Reviews*, vol. 151, p. 111530, Nov. 2021, doi: <https://doi.org/10.1016/j.rser.2021.111530>.
- [110] Amali Matharaarachchi et al., “Optimizing Generative AI Chatbots for Net-Zero Emissions Energy Internet-of-Things Infrastructure,” *Energies*, vol. 17, no. 8, pp. 1935–1935, Apr. 2024, doi: <https://doi.org/10.3390/en17081935>.
- [111] C. Wu et al., “Natural language processing for smart construction: Current status and future directions,” *Automation in Construction*, vol. 134, p. 104059, Feb. 2022, doi: <https://doi.org/10.1016/j.autcon.2021.104059>.
- [112] N. Tyagi and B. Bhushan, “Demystifying the Role of Natural Language Processing (NLP) in Smart City Applications: Background, Motivation, Recent Advances, and Future Research Directions,” *Wireless Personal Communications*, Mar. 2023, doi: <https://doi.org/10.1007/s11277-023-10312-8>.
- [113] Alaa Awad Abdellatif, Naram Mhaisen, A. Mohamed, Aiman Erbad, and Mohsen Guizani, “Reinforcement Learning for Intelligent Healthcare Systems: A Review of Challenges, Applications, and Open Research Issues,” *IEEE internet of things journal (Online)*, vol. 10, no. 24, pp. 21982–22007, Dec. 2023, doi: <https://doi.org/10.1109/jiot.2023.3288050>.
- [114] Y. Chen, S. Han, G. Chen, J. Yin, Kate Nana Wang, and J. Cao, “A deep reinforcement learning-based wireless body area network offloading optimization strategy for healthcare services,” *Health information science and systems*, vol. 11, no. 1, Jan. 2023, doi: <https://doi.org/10.1007/s13755-023-00212-3>.
- [115] Fisayo Sangoleye, J. Jao, K. Faris, Eirini Eleni Tsiropoulou, and Symeon Papavassiliou, “Reinforcement Learning-Based Demand Response Management in Smart Grid Systems With Prosumers,” *IEEE Systems Journal*, vol. 17, no. 2, pp. 1797–1807, Mar. 2023, doi: <https://doi.org/10.1109/jsyst.2023.3248320>.
- [116] L. Yun, D. Wang, and L. Li, “Explainable multi-agent deep reinforcement learning for real-time demand response towards sustainable manufacturing,” *Applied energy*, vol. 347, pp. 121324–121324, Oct. 2023, doi: <https://doi.org/10.1016/j.apenergy.2023.121324>.
- [117] Y. Wang, F. Shang, J. Lei, X. Zhu, H. Qin, and J. Wen, “Dual-attention assisted deep reinforcement learning algorithm for energy-efficient resource allocation in Industrial Internet of Things,” *Future Generation Computer Systems*, vol. 142, pp. 150–164, May 2023, doi: <https://doi.org/10.1016/j.future.2022.12.009>.
- [118] N. Deo and A. Anjankar, “Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India,” *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, vol. 15, no. 5, May 2023, doi: <https://doi.org/10.7759/cureus.39416>.
- [119] Z. Qin, Z. D. Xu, Q. C. Sun, P. Poovendran, and P. Balamurugan, “Investigation of Intelligent Substation Inspection Robot by Using Mobile Data,” *International Journal of Humanoid Robotics*, May 2022, doi: <https://doi.org/10.1142/s0219843622400035>.
- [120] Prasenjit Bhadra, S. Chakraborty, and S. Saha, “Cognitive IoT Meets Robotic Process Automation: The Unique Convergence Revolutionizing Digital Transformation in the Industry 4.0 Era,” *Smart Innovation, Systems and Technologies*, pp. 355–388, Jan. 2023, doi: https://doi.org/10.1007/978-981-19-8296-5_15.
- [121] F. Subhan et al., “AI-Enabled Wearable Medical Internet of Things in Healthcare System: A Survey,” *Applied Sciences*, vol. 13, no. 3, p. 1394, Jan. 2023, doi: <https://doi.org/10.3390/app13031394>.
- [122] M. Khalid, “Energy 4.0: AI-enabled digital transformation for sustainable power networks,” *Computers & industrial engineering*, pp. 110253–110253, May 2024, doi: <https://doi.org/10.1016/j.cie.2024.110253>.
- [123] M. Soori, B. Arezoo, and R. Dastres, “Internet of things for smart factories in industry 4.0, a review,” *Internet of Things and Cyber-Physical Systems*, vol. 3, pp. 192–204, 2023, doi: <https://doi.org/10.1016/j.iotcps.2023.04.006>.
- [124] H. Goswami and P. Kumar, “Is Artificial Intelligence a Helping Hand for the Future of Neurosurgery?,” *2021 5th International Conference on Information Systems and Computer Networks (ISCON)*, Oct. 2021, doi: <https://doi.org/10.1109/iscon52037.2021.9702473>.
- [125] T. Feng, “Applications of Artificial Intelligence to Diagnosis of Neurodegenerative Diseases,” *Studies in health technology and informatics*, Nov. 2023, doi: <https://doi.org/10.3233/shti230896>.
- [126] Z. Lu, I. Afridi, H. J. Kang, I. Ruchkin, and X. Zheng, “Surveying neuro-symbolic approaches for reliable artificial intelligence of things,”

Journal of Reliable Intelligent Environments, Jul. 2024, doi:
<https://doi.org/10.1007/s40860-024-00231-1>.

- [127] T. Kimura, "Virtual Teams: A Smart Literature Review of Four Decades of Research," *Human behavior and emerging technologies*, vol. 2024, pp. 1–20, Feb. 2024, doi:
<https://doi.org/10.1155/2024/8373370>.
- [128] R. Raman, V. K. Nair, and P. Nedungadi, "Discrepancies in Mapping Sustainable Development Goal 3 (Good Health and Well-Being) Research: A Comparative Analysis of Scopus and Dimensions Databases," *Sustainability*, vol. 15, no. 23, p. 16413, Jan. 2023, doi:
<https://doi.org/10.3390/su152316413>.



PREMA NEDUNGADI received a Ph.D. degree in Computer Science and Engineering from Amrita Vishwa Vidyapeetham, India. She is the Director of the Amrita Center for Research in Analytics, Technologies & Education (AmritaCREATE), Amrita University, and a Professor at the Amrita School of Computing, Amrita Vishwa Vidyapeetham, India. She is a recipient of the Digital India Award from the Ministry of Electronics and Information Technology, India, in the category of digital empowerment. She was a finalist in the U.S. \$7 million Barbara Bush Foundation Adult Literacy XPRIZE Competition.

SIMI SURENDRAN is an Assistant Professor (Sr. Gr.) and Vice-Chair, CSE(AI) at the School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri. She received her Ph. D. in Wireless Networks and Applications from Amrita Vishwa Vidyapeetham, India. She has an M.Tech degree in Wireless network and Applications from Amrita Vishwa Vidyapeetham, India, and a B.Tech degree in Computer Science and Engineering from Mahatma Gandhi University, India. She was also selected as a candidate for the Erasmus International Credit Mobility Program at the University of Trento, Italy, as part of the Ph.D. program. Her research interests are in networking in extreme environments, stochastic algorithms, predictive analytics, and reinforcement learning. In collaboration with the University of Trento, she has published papers in reputed journals and conferences in the field of wireless networks with a particular emphasis on predictive analytics.



KAI-YU TANG received his Ph.D. from National Chiao Tung University. He is an Associate Professor at the Graduate Institute of Library & Information Science, National Chung Hsing University, Taiwan. His research interests include electronic commerce, computational thinking, and social network analysis. His research papers have been published in Computers & Education, Electronic Commerce Research and Applications, Educational Technology Research and Development, the International Journal of Computer-Supported Collaborative Learning, Scientometrics, Telematics and Informatics, and the International Journal of Information Management.



RAGHU RAMAN received a Ph.D. degree in Management from Amrita Vishwa Vidyapeetham, India, and an M.B.A. degree from the Haas School of Business, UC Berkeley, Berkeley, CA, USA. He is currently the Dean of the School of Business, Amrita Vishwa Vidyapeetham. He has over 30 years of executive management experience at a variety of Fortune 500 companies and has been with Amrita Vishwa Vidyapeetham since its founding in 2003. He established the Center for Research in Analytics and Technologies for Education (CREATE), with over U.S. \$5 million in research funding. He was a recipient of the President of India Gold Medal in 1986. He serves on the Board of Directors for Amrita Technology Business Incubator. He is the Past Chair of the IEEE Education Society Chapter India.